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- (54) **PHYSICAL LAYER SECURITY**
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- (73) Assignee: **QDAComm LLC**, Raleigh, NC (US)
- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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H04W 12/037 (2021.01)
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CPC **H04W 12/037** (2021.01); **H04L 69/323**
(2013.01)
- (58) **Field of Classification Search**
CPC H04W 12/037; H04L 69/323
See application file for complete search history.

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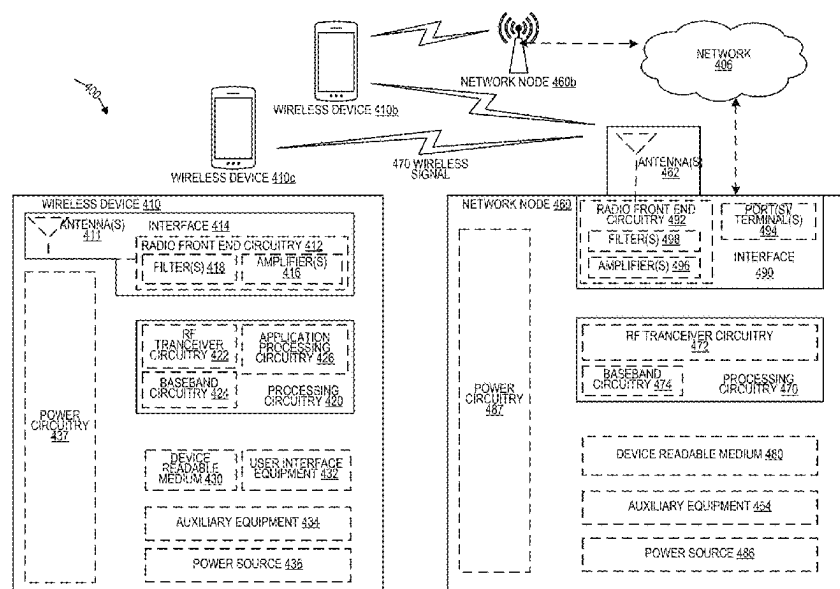
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(57) **ABSTRACT**

Systems and methods of performing physical layer security are presented. In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a certain physical layer security scheme and is associated with a quantized polar representation of the input signal sample.

30 Claims, 6 Drawing Sheets



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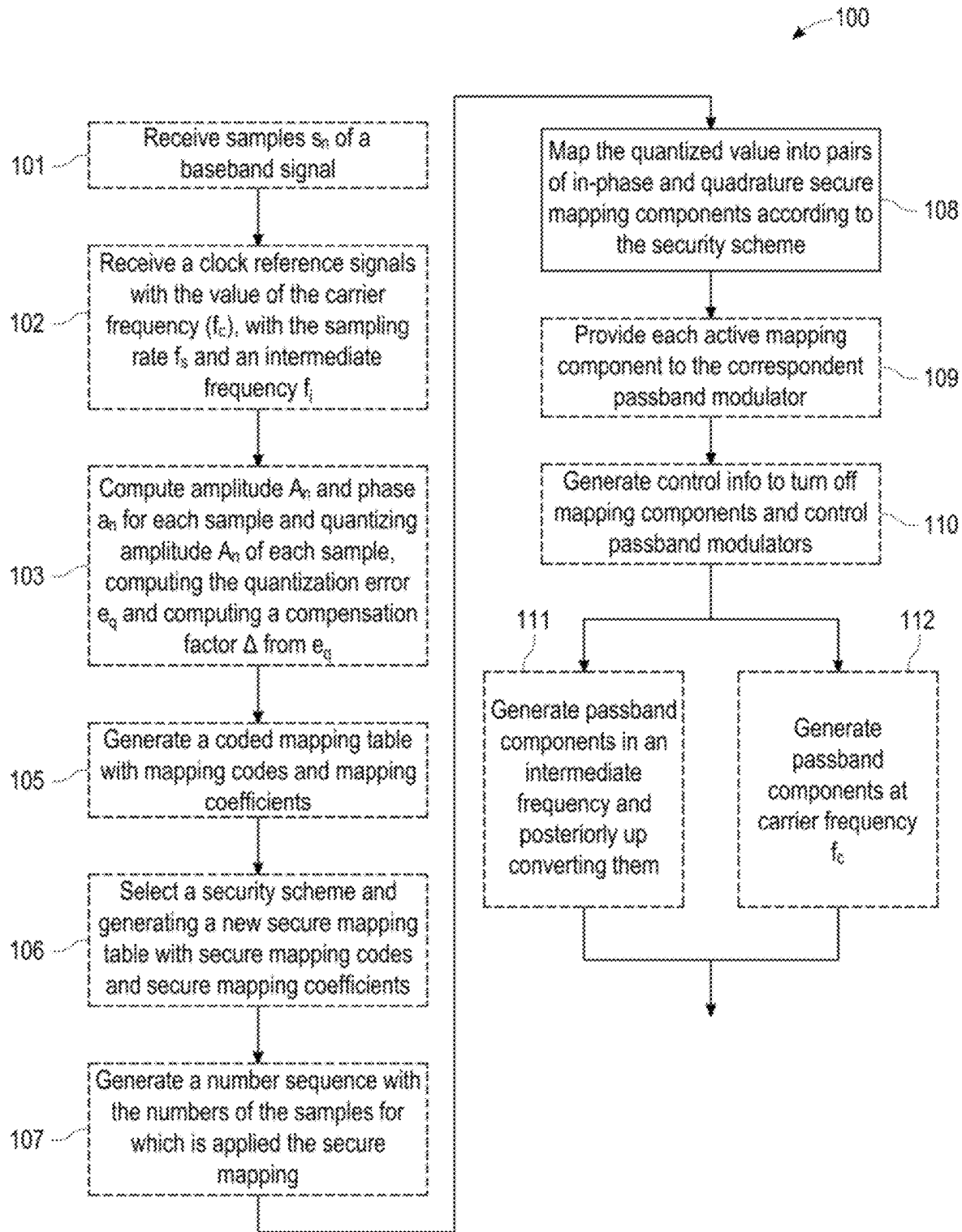


FIG. 1

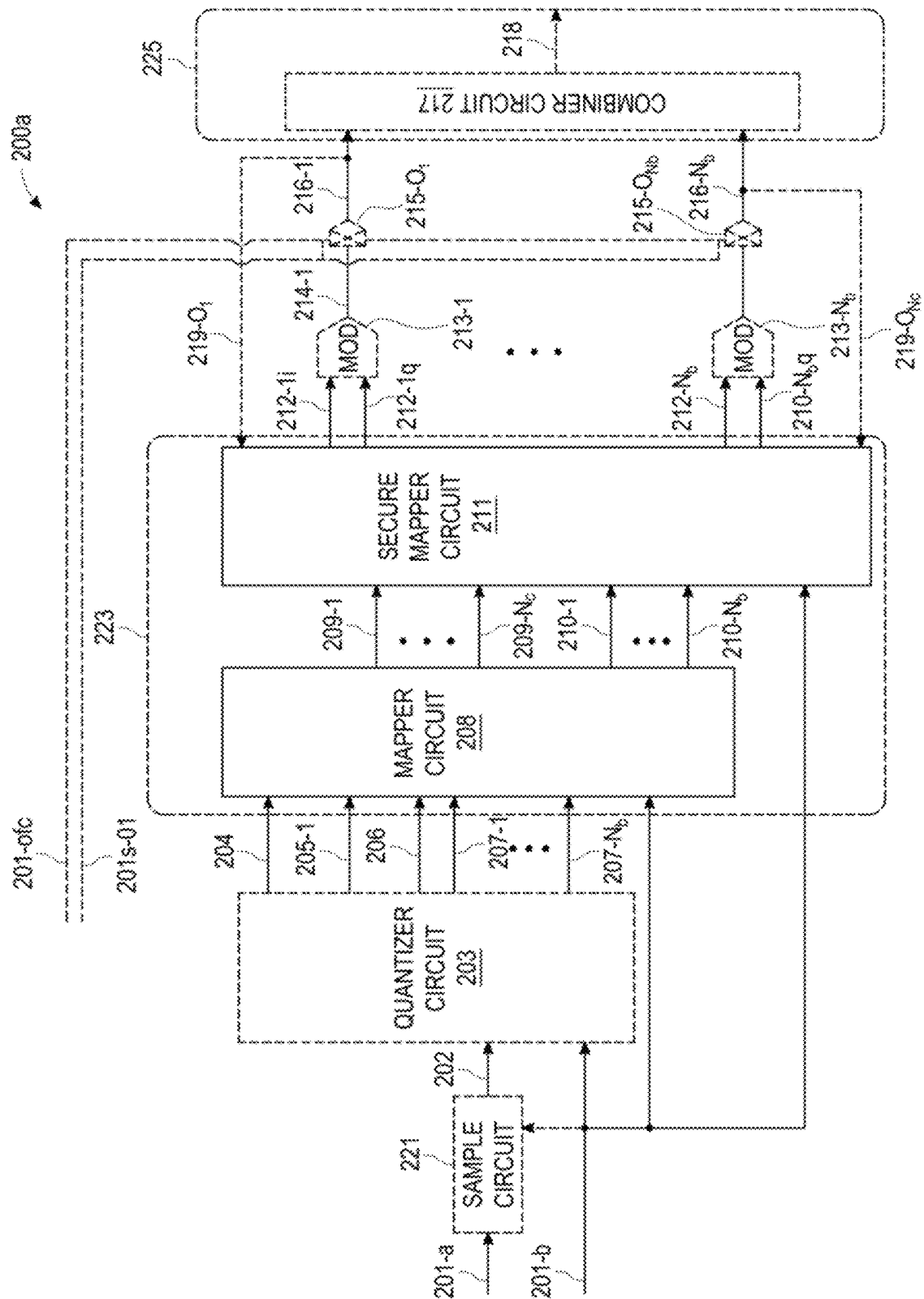


FIG. 2A

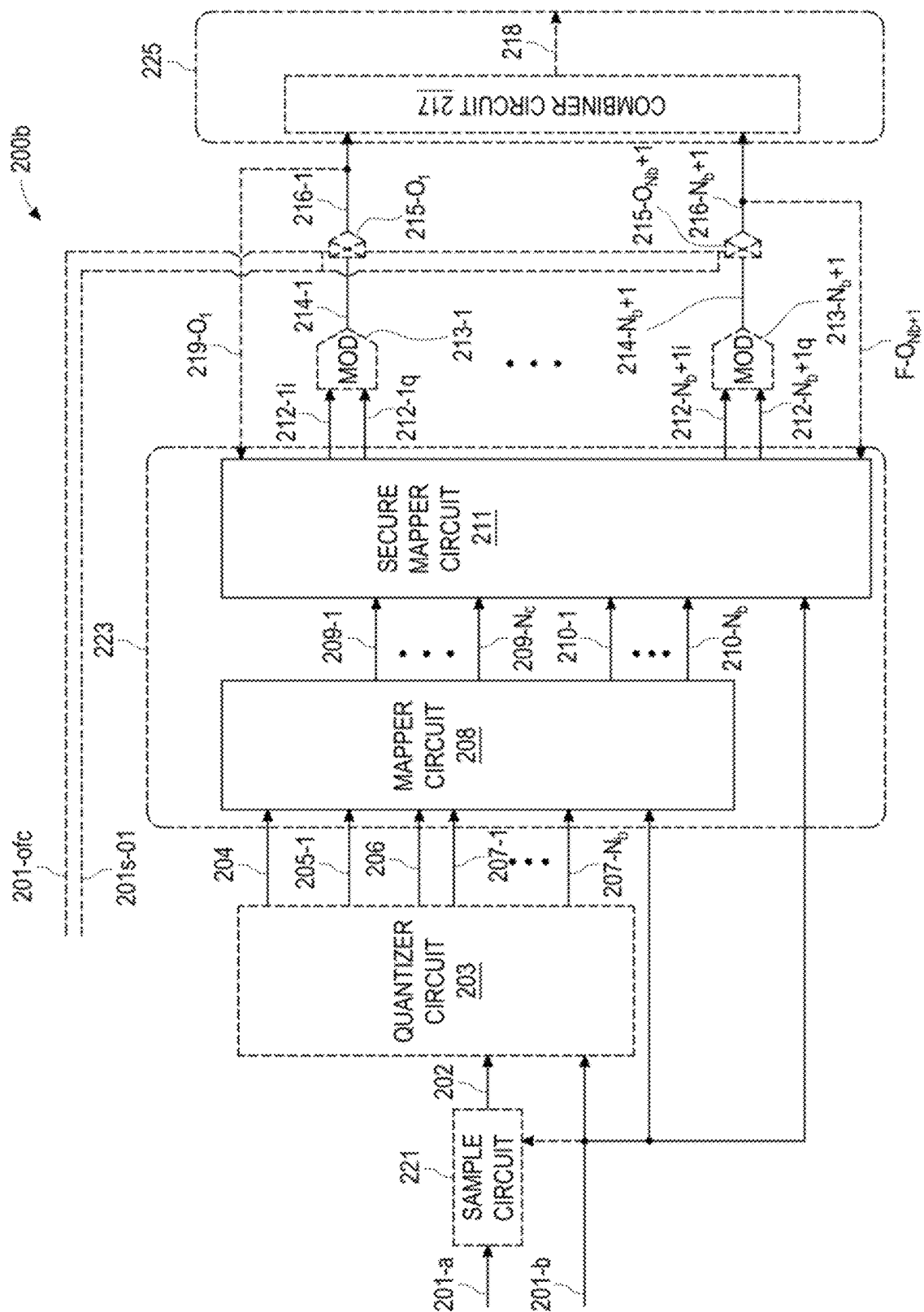


FIG. 2B

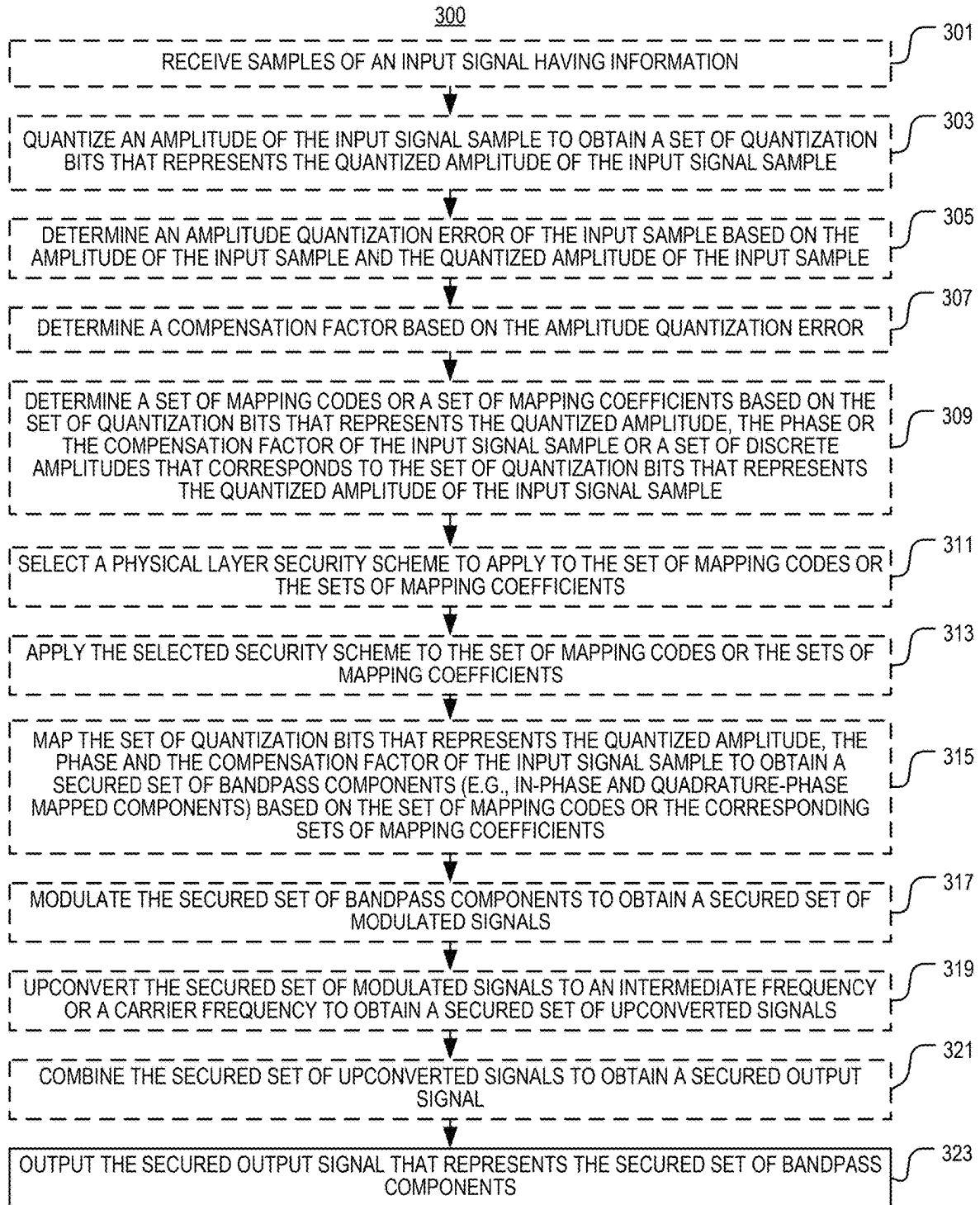
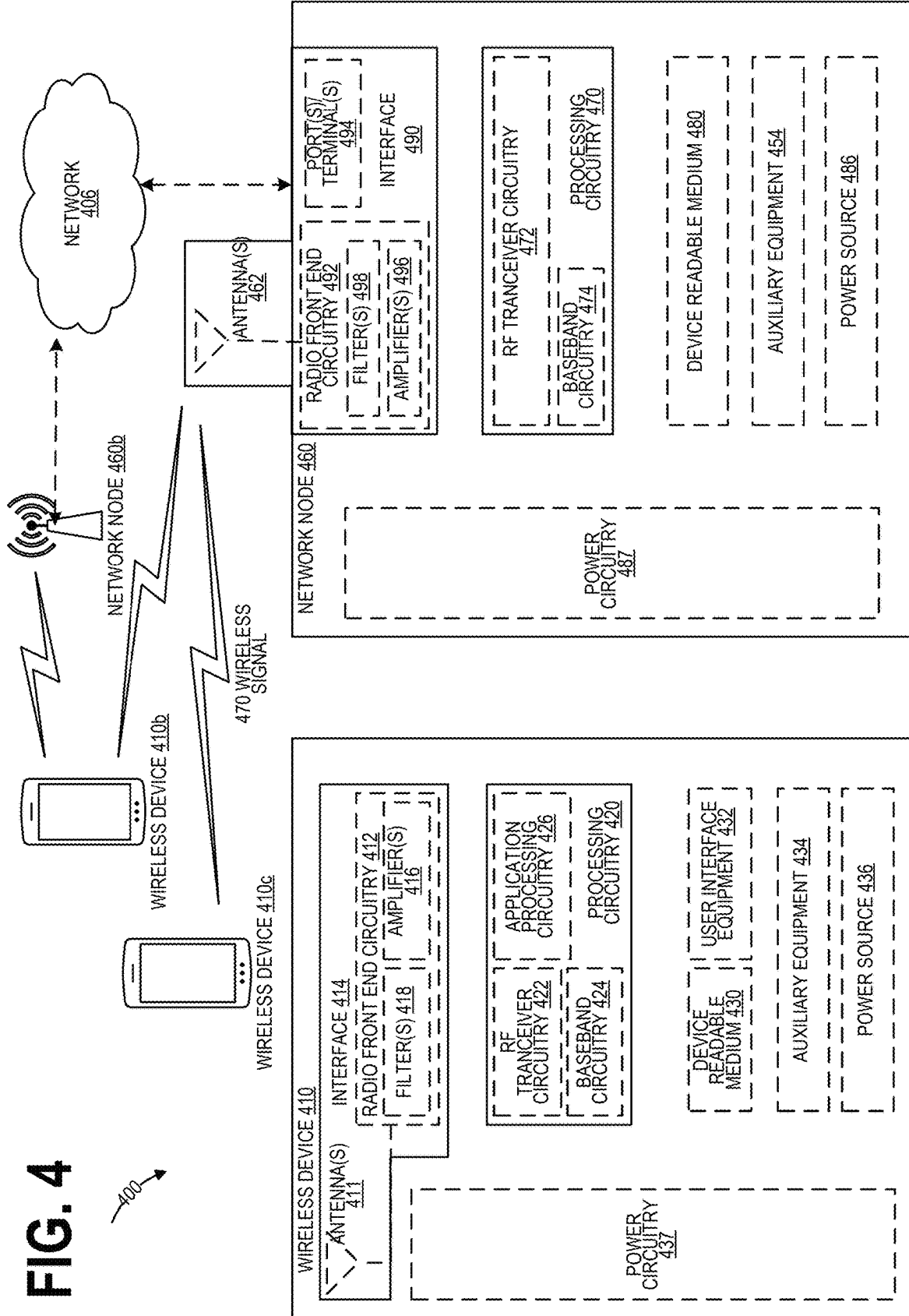


FIG. 3

FIG. 4



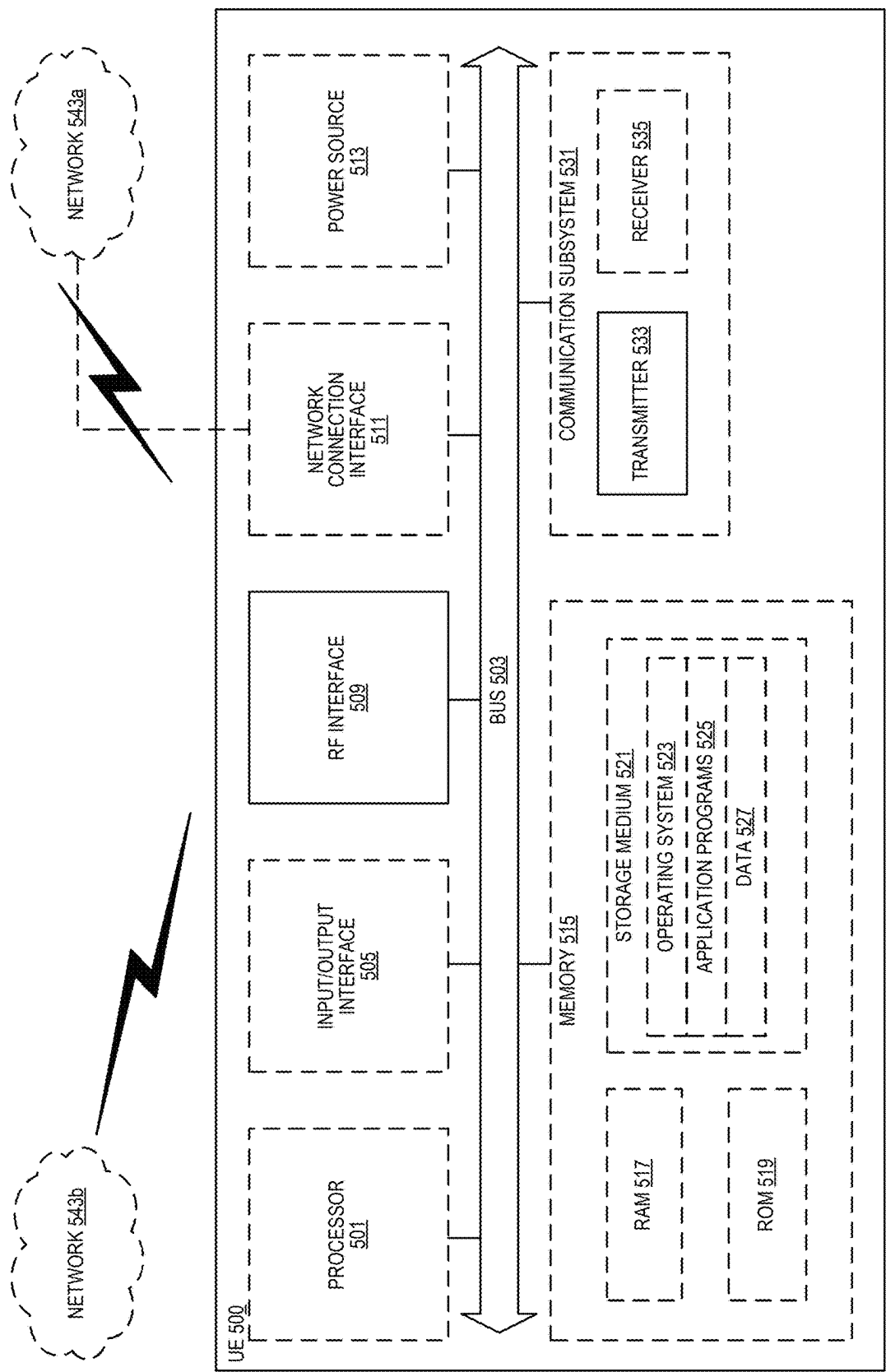


FIG. 5

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PHYSICAL LAYER SECURITY

FIELD OF THE DISCLOSURE

The present disclosure relates generally to physical layer security and more specifically to physical layer security such as that enabled by a coded decomposition of digital samples of time variant envelope signals into mapping components for energy optimization of transceiver front ends.

BACKGROUND

Wireless communications are prone to interception of wirelessly transmitted signals by unauthorized wireless devices. Due to the sharing nature of a wireless communication channel, other wireless devices that are present on the same wireless communication channel at transmission may have access to the transmitted data. Accordingly, there is a need for improved techniques to avoid unauthorized or unwanted interception of transmitted data on a wireless communication channel. In addition, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and embodiments, taken in conjunction with the accompanying figures and the foregoing technical field and background.

The Background section of this document is provided to place embodiments of the present disclosure in technological and operational context, to assist those of skill in the art in understanding their scope and utility. Unless explicitly identified as such, no statement herein is admitted to be prior art merely by its inclusion in the Background section.

BRIEF SUMMARY OF THE DISCLOSURE

The following presents a simplified summary of the disclosure in order to provide a basic understanding to those of skill in the art. This summary is not an extensive overview of the disclosure and is not intended to identify key/critical elements of embodiments of the disclosure or to delineate the scope of the disclosure. The sole purpose of this summary is to present some concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.

In this disclosure, a system of providing physical security with codified amplification is provided. According to one aspect, a system can receive samples $s_n = s(t_n) = s_{I,n} + js_{Q,n}$ of a baseband signal $s(t)$ at sampling instants t_n of the sampled in-phase and quadrature-phase components of the baseband signal $s(t)$, can receive an input clock signal associated with the sampling frequency $f_s = 1/T_s$ of the baseband signal $s(t)$, can receive a clock signal associated with an intermediate frequency f_i , and can receive a clock signal associated with the carrier frequency f_c of the output signal.

A quantizer circuitry can receive the samples of s_n of the baseband signal $s(t_n) = s_{I,n} + js_{Q,n}$, having sampling instants t_n , can compute the amplitude

$$A_n = |s_n| = \sqrt{s_{I,n}^2 + js_{Q,n}^2}$$

and the phase

$$\alpha_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right)$$

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of each sample using a quantization encoding table with N_b bits, a table with a finite set of quantization values and a table with discrete amplitudes of quantization components $a_i = \{a_1, a_2, \dots, a_{N_b}\}$, with $N_b = N_b$ or $N_b = N_b + 1$, in which the quantized value can be decomposed. The quantizer circuitry can generate the quantized amplitude $A_{qn} = A_n + e_q$ of each sample, where e_q denotes the quantization error and can generate a set of quantization bits and, based on the quantization error e_q , can compute a compensation factor Δ belonging to a finite discrete alphabet.

A mapper circuitry can receive the amplitude A_q , phase α_n and compensation factor Δ for each sample, and together with the quantization table, can generate a coded mapping table with mapping codes and corresponding mapping coefficients. A secure mapper circuitry can receive the coded mapping table and the quantization table and can select the security scheme to be applied to the mapping codes and the mapping coefficients needed to generate a secure mapping table to be used in generation of bandpass components. The secure mapper circuitry can generate numerical sequences having the positions of samples in which secure mapping tables should be applied. It should be noted that the secure mapper circuitry can be configured to have numerical sequences stored in a memory element such as a look up table (LUT), read only memory (ROM), random access memory (RAM), flash memory, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), cache memory, register memory, or the like. The secure mapper circuitry can generate the pairs of in-phase and quadrature-phase mapping components and can provide the active pair of mapping components to the corresponding bandpass modulator.

The secure mapper can apply the secure mapping table with mapping coefficients C_k' where $k=1, \dots, N_b$, with $N_b = N_b$ or $N_b = N_b + 1$, to create secure mapping tables with new mapping coefficients C_k' needed to achieve security on mapping the quantized value into a maximum number $N' \leq N_b$ of active pairs of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ components related to $N' \leq N_b$ active mapping components given by $C_k' \cos(\alpha_n)$ and $C_k' \sin(\alpha_n)$.

According to another aspect, the maximum number of possible bandpass components and mapping coefficients C_k' is N_b , and coefficients C_k' have values from N_b discrete alphabets.

According to another aspect, the maximum number of possible bandpass components or mapping coefficients C_k' is $N_b + 1$ for N_b quantization bits and coefficients C_k' and C_k' can represent values from $N_b + 1$ discrete alphabets. Secure mapping coefficients can be applied by bandpass modulators to generate the bandpass components required by the secure mapper circuit. The set of active bandpass components can be generated at carrier frequency f_c .

According to another aspect, a set of memory elements (e.g., LUTs) can include the mapping table, the secure mapping table, the security scheme, the quantization encoding table, the sets of discrete amplitudes a_i , the set of quantization values, the like, or any combination thereof.

According to another aspect, a security scheme can be based on permutations of mapping codes and corresponding mapping coefficients.

According to another aspect, a security scheme can be based on phase rotations of mapping coefficients associated with mapping codes that can affect all or a subset of the mapping codes.

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According to another aspect, a security scheme can be based on multiplying a scale factor and mapping coefficients associated with mapping codes that can affect all or a sub-set of the mapping codes.

According to another aspect, a security scheme can be based on puncturing mapping codes.

According to another aspect, a security scheme can be based on any combination of permutations, puncturing, phase rotations and scale factor multiplication.

According to another aspect, any of the security schemes disclosed herein can be combined or applied at the same time or applied at separate times to different samples or subsets of samples.

According to another aspect, any of the security schemes disclosed herein can be combined with interleaving techniques affecting the sequence of samples of the input signal.

According to another aspect, the secure mapping rule applied to each quantized value can be applied by a secure mapper circuitry to generate control information to enable bandpass modulators associated with the mapping components.

According to another aspect, the bandpass modulator circuitry can be configured as an in-phase/quadrature-phase (I/Q) digital-to-analog converter (I/Q DAC) or configured as an I/Q modulator.

According to another aspect, the secure mapper circuitry can provide information to control the phase of bandpass components generated by a set of bandpass modulators (I/Q modulators or I/Q DACs).

According to another aspect, the set of bandpass modulators can generate active bandpass components at the intermediate frequency f_i and a set of mixer circuitry can upconvert the active bandpass components to the carrier frequency f_c .

According to another aspect, the method and apparatus for performing physical security with codified amplification includes a digital circuitry operable to compute A_n and α_n , quantize A_n , perform the mapping encoding, define the secure mapping rule, perform the secure mapping of the quantized value into bandpass components and generate random permutation patterns, phase rotations or puncturing patterns.

According to another aspect, the method and apparatus for performing physical security with codified amplification can be implemented with analog or digital controls, including hardware, firmware or software.

According to another aspect, the method and apparatus for performing physical security with codified amplification can be implemented with analog components or with a combination of analog components and digital components.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the disclosure are shown. However, this disclosure should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements throughout.

FIG. 1 illustrates one embodiment of a method of providing physical layer security in accordance with various aspects as described herein.

FIG. 2A illustrates one embodiment of a system of physical layer security in accordance with various aspects as

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described herein. FIG. 2B illustrates another embodiment of a system of physical layer security in accordance with various aspects as described herein.

FIG. 3 illustrates another embodiment of a method of providing physical layer security in accordance with various aspects as described herein.

FIG. 4 illustrates one embodiment of a wireless communication network in accordance with various aspects as described herein.

FIG. 5 illustrates one embodiment of an electronic device in accordance with various aspects as described herein.

DETAILED DESCRIPTION

For simplicity and illustrative purposes, the present disclosure is described by referring mainly to exemplary embodiments thereof. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be readily apparent to one of ordinary skill in the art that the present disclosure may be practiced without limitation to these specific details.

Common security techniques rely on encryption schemes from higher layers, such as key cryptography protocols. Besides the possible vulnerability associated with the increased computational capacity of eavesdroppers, these techniques achieve security at the expense of spectral and energy inefficiencies of the transmission system. Another problem with these security techniques is the high data throughput, resulting in undesirable cryptographic or security coding with high overheads that may compromise overall efficiency and increase latency of the system.

In the present disclosure, both low complexity and energy efficient PLS can be achieved, exploring the properties of coded mapping associated with signal's envelope decomposition in codified amplification. For instance, the coding rule applied to the mapping process of quantized value of signal envelope samples into bandpass components can be securely changed. Mapping codes can be varied and can be associated with mapping coefficients that when combined can generate each bandpass mapping component. Hence, mapping code changes such as permutations or puncturing of mapping codes can enable physical layer security. Further, mapping coefficients can be modified such as manipulation of the phase or amplitude of the mapping coefficients, which can further increase the degrees of freedom of possible physical layer security implementations.

In the present disclosure, the quantized digital amplification in the decomposition of quantized sample values into constant envelope components can apply a coded mapping rule having different mapping codes that are associated with subsets of mapping coefficients. Hence, changes to the mapping codes and the corresponding mapping coefficients can be utilized to enable physical layer security, without additional complexity. This type of physical layer security remains undetectable as the transmission spectral mask still conforms to any standards requirements and the statistical characteristics of the signal envelope also remains Gaussian. Any transmission overhead is also avoided as well as any dependency on transmission channel statistics. Moreover, this quantized digital amplification can be applied along with a fully quantized digital amplification stage having a set of parallel amplifiers for improved efficiency or can be used with common power amplifiers employed in modern radio frequency front ends.

In one exemplary embodiment, a digital signal processing implementation can generate quantized punctured bandpass

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components that optimize energy efficiency on power amplification and power combination. A first mapper component (e.g., software, firmware, hardware) can decompose a quantized input signal sample into a sum of bandpass components according to mapping codes and can provide a reference for a second mapper component (e.g., software, firmware, hardware) to perform a secure mapping process based on a new secure mapping rule. A secure mapping rule can include generating permutations of the mapping codes, applying a phase rotation or a scale factor to the mapping coefficients, puncturing the mapping codes, the like, or any combination thereof. Further, a secure mapping rule can combine the several techniques and may change over the time or between subsets of the samples of a data block. The application of a secure mapping rule may not increase the complexity of a hardware-based transceiver as digital signal processing can be applied to perform the quantized digital amplification. Also, the quantized digital amplification can be applied along with a multi-amplifier power amplification stage followed by a combiner circuitry, optimized for the signal decomposition associated with the quantized digital amplification for improved energy efficiency.

FIG. 1 illustrates one embodiment of a method **100** of providing physical layer security in accordance with various aspects as described herein. In FIG. 1, the method **100** can start, for instance, at block **101** where it can include receiving the samples s_n of a baseband signal $s(t)$. In one example, the samples $s_n = s(t_n) = s_{I,n} + js_{Q,n}$ of a baseband signal $s(t)$ are quantized and decomposed into several analog bandpass components using a secure mapping rule. In another example, the input signal is a baseband signal which is sampled to generate the samples to be converted posteriorly into bandpass components. In yet another example, the samples $s_{I,n}$ and $s_{Q,n}$ of the respective in-phase and quadrature-phase components of a baseband signal are received. In yet another example, a baseband signal is received and sampled by a sampling circuitry or by a sample and hold (S/H) circuitry to generate the samples of the input signal. In yet another embodiment, the in-phase and quadrature-phase components of a baseband signal are received. At block **102**, the method **100** can include receiving a clock reference with a sampling rate f_s , receiving a clock reference with a carrier frequency f_c , and receiving a clock reference with an intermediate frequency f_i . At block **103**, the method **100** can include computing the amplitude A_n and the phase α_n for each sample, quantizing A_n as A_q using N_b quantization bits from a quantization table, computing the quantization error e_q , and computing a compensation factor Δ from e_q . At block **104**, the method **100** can include providing to a mapper circuitry the values of A_q , α_n , Δ , and the quantization encoded table. At block **105**, the method **100** can include generating a coded mapping table with mapping codes and corresponding mapping coefficients. At block **106**, the method **100** can include receiving the coded mapping table and the coded quantization table and selecting the security scheme to be applied to the mapping codes and the mapping coefficients of a secure mapping table to be applied in generation of the bandpass components. At block **107**, the method **100** can include generating a secure mapping table and numerical sequences having the positions of samples in which the secure mapping table will be applied in generation of the bandpass components. At block **108**, the method **100** can include generating the pairs of in-phase and quadrature-phase mapping components. At block **109**, the method **100** can include providing each active pair of mapping components to the corresponding bandpass modulator circuitry. At block **110**, the method **100** can include generating control

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information to enable or disable the bandpass modulator circuitry. At block **111**, the method **100** can include upconverting the bandpass components to an intermediate frequency f_i . At block **112**, the method **100** can include upconverting the bandpass components or the intermediate frequency components f_i to a carrier frequency f_c .

In another embodiment, the method steps described by blocks **103** to **109** can be performed in a single method step.

In another embodiment, the coded mapping table can be stored in a memory element (e.g., LUT) and as such, the method step described by block **105** is not required.

In another embodiment, as understood by a skilled artisan, based on the subject matter of the current disclosure, the method steps described by blocks **101** to **109** can be performed by memory elements (e.g., LUTs) having the quantization values, the quantization bits table, the coded mapping table, the secure mapping table, the numerical sequences having the positions of samples in which the secure mapping table can be applied to generate secure pairs of mapping components and control information.

In another embodiment, the security scheme can be changed between successive samples or between successive blocks of samples.

FIG. 2A illustrates one embodiment of a system **200a** of physical layer security with codified amplification in accordance with various aspects as described herein. In FIG. 2A, the system **200a** implements the method **100** of FIG. 1 with $N_b=3$ or $N_b=4$ quantization bits and the same number of mapping components. The number of mapping codes is 2^{N_b} , which is associated with N_b sets of mapping coefficients, each mapping coefficient with 2^{N_b} possible values. The system **200a** can receive an input signal **201-a** having the time samples $s_n = s(t_n)$ of a baseband signal $s(t)$. The system **200a** can also receive a clock reference signal **201-b** having the sampling frequency f_s . Additionally or alternatively, the system **200a** can receive the clock reference signal **201-s-o1** having an intermediate frequency f_i to be applied to mixer circuitry and a reference signal clock **201-ofc** having a carrier frequency f_c . A skilled artisan will readily recognize that the choice of the clock reference signal can be made according to the bandwidth of the input signal, the desired frequency for the output signal, and a frequency shift of upconverting bandpass components.

In another embodiment, the input signal **201-a** can include the samples $s_{I,n}$ and $s_{Q,n}$ of the respective in-phase and quadrature-phase components of a baseband signal.

In another embodiment, the input signal **201-a** can be a baseband signal $s(t)$, which can be sampled by sampling circuitry **221**, to generate the time samples $s_n = s(t_n)$.

The scope of this disclosure can include the application of other clock references or other input signals, and the implementation of such variations will be readily apparent to a skilled artisan. The clock signal **201-b** can be a clock reference signal having the sampling frequency f_s and can be input to the sampling circuitry **221**, the quantizer circuitry **203**, the mapper circuitry **208**, the secure mapper circuitry **211**, the set of bandpass modulator circuitry **213**- $\{1, \dots, N_b+1\}$, the combiner circuitry **217**, or the like.

The quantizer circuitry **203**, for each sample s_n , can determine the amplitude

$$A_n = |s_n| = \sqrt{s_{I,n}^2 + s_{Q,n}^2}$$

and the phase

$$\alpha_n = \tan^{-1} \left(\frac{s_{Q,n}}{s_{I,n}} \right).$$

The quantizer circuitry **203** can apply a quantization encoding table having N_b quantization bits, can apply a table having the quantization values, and a table having the set of discrete amplitudes of quantization components $a_i = \{a_1, a_2, \dots, a_{N_b}\}$ in which any quantized value can be decomposed. For each sample s_n , the quantizer circuitry **203** can quantize the amplitude A_n , can generate the quantized amplitude $A_{qn} = A_n + e_q$ and the set of quantization amplitude bits **207**- $\{1, \dots, N_b\}$, and can determine a quantization error compensation factor Δ belonging to a finite discrete alphabet based on the quantization error e_q . Further, the quantizer circuitry **203** can generate and send the quantization amplitude table **204** to the mapper circuitry **208**. In addition, the quantizer circuitry **203** can send, to the mapper circuitry **208**, the quantized amplitude bits **207**- $\{1, \dots, N_b\}$, the quantized amplitude A_q **205-1**, the phase α_n **205-2** and the amplitude quantization error Δ **205-3**, and the discrete amplitudes of quantization components a_i **206**.

In FIG. **2A**, the mapper circuitry **208** can determine a coded mapping table having mapping codes and corresponding mapping coefficients. The mapper circuitry **208** can apply the quantization table to define the set of mapping codes $M_i = \{M_0; \dots; M_{N_c}\}$, with $N_c = 2^{N_b} - 1$, which can be associated with the sets of mapping coefficients $C_{N_b}^{(0)} \dots C_1^{(0)}$, $\dots$, $C_{N_b}^{(N_c)} \dots C_1^{(N_c)}$ and can be defined in terms of the mapping coefficients $C_k = \{C_k^{(1)}; \dots; C_k^{(N_c)}\}$, with $k=1, \dots, N_b$. For example, with $N_b=3$ quantization bits, the mapper circuitry **208** can define an encoded mapping as shown in Table I below, with eight (8) mapping codes $M_i = \{M_0; \dots; M_7\}$ directly related to the sets of mapping coefficients $C_3^{(0)}C_2^{(0)}C_1^{(0)}$, $C_3^{(1)}C_2^{(1)}C_1^{(1)}$, $\dots$, $C_3^{(7)}C_2^{(7)}C_1^{(7)}$.

TABLE I

Mapping Coefficients and Mapping Codes for $N_b = 3$			
C_3	C_2	C_0	Mapping Code M_i
$C_3^{(0)}$	$C_2^{(0)}$	$C_1^{(0)}$	M_0
$C_3^{(1)}$	$C_2^{(1)}$	$C_1^{(1)}$	M_1
$C_3^{(2)}$	$C_2^{(2)}$	$C_1^{(2)}$	M_2
$C_3^{(3)}$	$C_2^{(3)}$	$C_1^{(3)}$	M_3
$C_3^{(4)}$	$C_2^{(4)}$	$C_1^{(4)}$	M_4
$C_3^{(5)}$	$C_2^{(5)}$	$C_1^{(5)}$	M_5
$C_3^{(6)}$	$C_2^{(6)}$	$C_1^{(6)}$	M_6
$C_3^{(7)}$	$C_2^{(7)}$	$C_1^{(7)}$	M_7

C_1 mapping coefficients can represent values from a set of amplitudes $C_1^{(i)}$, $i=0, \dots, 7$ that belongs to a discrete alphabet A_1 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 or the amplitude quantization error Δ . C_2 mapping coefficients can represent values from a set of amplitudes C_2 , $i=0, \dots, 7$ that belongs to a discrete alphabet A_2 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , or the amplitude quantization error Δ . C_3 mapping coefficients can represent values from a set of amplitudes $C_3^{(i)}$, $i=0, \dots, 7$ that belongs to a discrete alphabet A_3 with a null value and other values defined as a

function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the third quantization component a_3 , or the amplitude quantization error Δ .

In another exemplary embodiment with $N_b=4$ quantization bits and $N_b = N_b$ possible bandpass components the mapper circuitry **208** can apply the quantization bits and the quantization table **206** to define an encoded mapping table with mapping codes $M_i = \{M_0; \dots; M_{15}\}$ directly related to the sets of mapping coefficients $C_4^{(0)}C_3^{(0)}C_2^{(0)}C_1^{(0)}$, $C_4^{(1)}C_3^{(1)}C_2^{(1)}C_1^{(1)}$, $\dots$, $C_4^{(15)}C_3^{(15)}C_2^{(15)}C_1^{(15)}$. C_1 mapping coefficients can represent values from a set of amplitudes $C_1^{(i)}$, $i=0, \dots, 15$ that belong to a discrete alphabet A_1 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , or the amplitude quantization error Δ . C_2 mapping coefficients can represent values from a set of amplitudes $C_2^{(i)}$, $i=0, \dots, 15$ that belong to a discrete alphabet A_2 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , or the amplitude quantization error Δ . C_3 mapping coefficients can represent values from a set of amplitudes $C_3^{(i)}$, $i=0, \dots, 15$ that belong to a discrete alphabet A_3 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , or the amplitude quantization error Δ . C_4 mapping coefficients can represent values from a set of amplitudes $C_4^{(i)}$, $i=0, \dots, 15$ that belong to a discrete alphabet A_4 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , or the amplitude quantization error Δ .

From the mapper circuitry **208**, the secure mapper circuitry **211** can receive the mapping codes **209**- $\{1, \dots, N_c\}$ and the mapping coefficients **210**- $\{1, \dots, N_b\}$ and can select the security scheme to apply in determining the updated secure mapping codes and the corresponding mapping coefficients. The secure mapper circuitry **211** can apply the security scheme to determine the secure mapping coefficients C_k' where $k=1, \dots, N_b$, and can generate secure mapping tables with new mapping coefficients C_k' required to achieve security in mapping the quantized amplitude value into a maximum number $N' \leq N_b$, of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ mapping components among the N_b possible values **212**- $\{1, \dots, N_{b,i}\}$, **212**- $\{1, \dots, N_{b,q}\}$, related to $N' \leq N_b$, active mapping components given by $C_k' \cos(\alpha_n)$ and $C_k' \sin(\alpha_n)$, respectively.

In another embodiment, a security scheme can include permutations between mapping codes, phase rotations of mapping coefficients, puncturing of mapping codes and coefficients, scale factor multiplication of mapping coefficients, the like, or any combination thereof. A secure mapper circuitry can generate a secure mapping table with secure mapping codes and corresponding mapping coefficients to be applied to the mapping of quantized values and can generate the number sequence with the samples in which the secure mapping rule should be applied.

In another embodiment, with permutations of mapping codes, a permutation array $P = \{P_0, P_1, \dots, P_{N_c}\}$, being the values adopted to define a set of permuted mapping codes $M_i' = \{M_{P_0}; M_{P_1}; \dots; M_{P_{N_c}}\}$, in which mapping codes are associated with the corresponding sets of secure mapping coefficients $C_{N_b}^{(P_i)}, C_{N_b-1}^{(P_i)}, \dots, C_1^{(P_i)}$, with $i=0, \dots, N_c$.

In one exemplary embodiment, with $N_b=3$ and $N_c=8$ mapping codes, and for a permutation array of $P = \{P_0,$

$P_1, \dots, P_7\}$, results in a new permuted set of mapping codes $M_i'=\{M_{P_0}; M_{P_1}; \dots; M_{P_7}\}$, which are associated with the corresponding sets of mapping coefficients given by $C_3^{(P_0)}C_2^{(P_0)}C_1^{(P_0)}$, $C_3^{(P_1)}C_2^{(P_1)}C_1^{(P_1)}$, $\dots$, $C_3^{(P_7)}C_2^{(P_7)}C_1^{(P_7)}$. For instance, a permutation array of $P=\{7, 0, 6, 1, 4, 2, 5, 3\}$ results in Table II below.

TABLE II

Example of Mapping Table After Permutation of Mapping Codes			
C_3	C_2	C_1	Mapping Code M_i
$C_3^{(p0)}$	$C_2^{(p0)}$	$C_1^{(p0)}$	$M'_0 = M_{P_0} = M_7$
$C_3^{(p1)}$	$C_2^{(p1)}$	$C_1^{(p1)}$	$M'_1 = M_{P_1} = M_0$
$C_3^{(p2)}$	$C_2^{(p2)}$	$C_1^{(p2)}$	$M'_2 = M_{P_2} = M_6$
$C_3^{(p3)}$	$C_2^{(p3)}$	$C_1^{(p3)}$	$M'_3 = M_{P_3} = M_1$
$C_3^{(p4)}$	$C_2^{(p4)}$	$C_1^{(p4)}$	$M'_4 = M_{P_4} = M_4$
$C_3^{(p5)}$	$C_2^{(p5)}$	$C_1^{(p5)}$	$M'_5 = M_{P_5} = M_2$
$C_3^{(p6)}$	$C_2^{(p6)}$	$C_1^{(p6)}$	$M'_6 = M_{P_6} = M_5$
$C_3^{(p7)}$	$C_2^{(p7)}$	$C_1^{(p7)}$	$M'_7 = M_{P_7} = M_3$

With eight (8) mapping codes, the number of possible permutations is forty thousand, three hundred and twenty (40,320) and permutations can change dynamically over each data block or within a data block according to a predefined pattern.

In another exemplary embodiment, with $N_b=4$ and $N_c=16$ mapping codes, for a permutation array $P=\{P_0, P_1, \dots, P_{15}\}$, results in a new permuted set of mapping codes $M_i'=\{M_{P_0}; M_{P_1}; \dots; M_{P_{15}}\}$, which are associated with the corresponding sets of mapping coefficients given by $C_4^{(0)}C_3^{(0)}C_2^{(0)}C_1^{(0)}$, $C_4^{(1)}C_3^{(1)}C_2^{(1)}C_1^{(1)}$, $\dots$, $C_4^{(15)}C_3^{(15)}C_2^{(15)}C_1^{(15)}$.

In another embodiment, the secure mapper circuitry can apply a puncturing array $Pu=\{pu_1, \dots, pu_{N_c}\}$, where each put represents a value of one (1) without puncturing or a value of X with puncturing, with the puncturing pattern to be applied to puncture mapping codes $M_i=\{M_0; M_1; \dots; M_{N_c}\}$. For instance, with $N_b=3$ and a puncturing rule of $Pu=\{1, X, 1, 1, X, 1, 1\}$, results in the set of punctured mapping codes $M_i'=\{M_0; M_1; \dots; M_7\}$, directly related to the sets of mapping coefficients $C_3^{(0)}C_2^{(0)}C_1^{(0)}$, X , $C_3^{(2)}C_2^{(2)}C_1^{(2)}$, X , $C_3^{(3)}C_2^{(3)}C_1^{(3)}$, X , $C_3^{(5)}C_2^{(5)}C_1^{(5)}$, $C_3^{(6)}C_2^{(6)}C_1^{(6)}$, $C_3^{(7)}C_2^{(7)}C_1^{(7)}$.

In another exemplary embodiment, with $N_b=4$ quantization bits and $N_b=N_b$ possible bandpass components for a puncturing rule $Pu=\{1, X, 1, 1, X, 1, 1, 1, 1, X, 1, 1, 1, X, 1\}$, results in the set of mapping codes $M_i'=\{M_0; X; M_2; M_3; X; M_5; \dots; M_9; X; M_{11}; M_{12}; M_{13}; X; M_{15}\}$, directly related to the sets of mapping coefficients: $C_4^{(0)}C_3^{(0)}C_2^{(0)}C_1^{(0)}$, X , $C_4^{(2)}C_3^{(2)}C_2^{(2)}C_1^{(2)}$, X , $C_4^{(3)}C_3^{(3)}C_2^{(3)}C_1^{(3)}$, X , $C_4^{(5)}C_3^{(5)}C_2^{(5)}C_1^{(5)}$, $\dots$, $C_4^{(9)}C_3^{(9)}C_2^{(9)}C_1^{(9)}$, X , $C_4^{(11)}C_3^{(11)}C_2^{(11)}C_1^{(11)}$, $\dots$, $C_4^{(13)}C_3^{(13)}C_2^{(13)}C_1^{(13)}$, X , $C_4^{(15)}C_3^{(15)}C_2^{(15)}C_1^{(15)}$.

In another embodiment, the puncturing rule can be constant over the entire data block or may differ for different subsets of samples within the data block.

In another embodiment, mapping codes can be punctured according to a predetermined rule.

In another embodiment, the puncturing rule can be applied as a random phase rotation $C_k^{(i)}=e^{j\alpha}C_k^{(i)}$ to a sub-set of mapping codes and corresponding mapping coefficients, with α denoting the phase rotation value. A phase rotation can affect all mapping codes or a subset of mapping codes. A phase rotation value can be predetermined, or the phase rotation may change over the data block or samples of the envelope according to a predetermined rule. This corre-

sponds to a phase rotation of the values of the alphabets A_k associated with the sets of possible values of mapping coefficients.

In another embodiment, a phase rotation can be applied to only a set of mapping coefficients $C_k^{(i)}=e^{j\alpha}C_k^{(i)}$, for $i=1, \dots, i_n$ and $k=1, \dots, N'$, with $N'<N_b$, with only some alphabets of possible values of mapping coefficients being affected by phase rotation.

In another embodiment, a scale factor can be applied to a limited set of mapping coefficients $C_{ks}^{(i)}=s_kC_{ks}^{(i)}$, for $i=1, \dots, i_n$ and $k_s=ks_1, ks_2, \dots, N'$, with $N'<N_b$.

In another embodiment, for a security scheme based on permutations of mapping codes, any combination of puncturing, phase rotations and multiplication of mapping coefficients by a scale factor can be applied to different subsets of samples.

In FIG. 2A, the secure mapper 211 can generate a set of $N' \leq N_b$ active pairs of in-phase and quadrature-phase components related to $N' \leq N_b$ mapping components, based on a secure mapping table such as stored in a memory element (e.g., LUT) that contains the mapping coefficient values of the secure mapping rule.

In another embodiment, the secure mapping table can also include the quantization bits, the sets of discrete amplitudes a_i and the set of quantized values.

In another embodiment, the mapping table, the secure mapping table, the quantization encoding table, the quantization values table and quantization amplitude components can be previously stored in memory elements (e.g., LUTs) that are available to the quantizer circuitry, the mapper, or the secure mapper circuit.

In another embodiment, memory elements (e.g., LUTs) can include the permutation rules, the puncturing rules, the phase rotations rules to be applied in the definition of the secure mapping tables that are also previously stored in those memory elements.

In another embodiment, the secure mapper circuitry 211 can generate numerical sequences having the positions of samples in which secure mapping tables are applied, or numerical sequences are previously stored in memory elements (e.g., LUTs).

In another embodiment, sets of numerical sequences with the positions of samples affected by security schemes can be previously stored in LUTs.

In FIG. 2A, the secure mapper 211 can output $N' \leq N_b$ active pairs of in-phase and quadrature-phase mapping components $212-\{1, \dots, N_{bi}\}$, $212-\{1, \dots, N_{bi}\}$, to a set of N' bandpass modulator circuitry $213-\{1, \dots, N_b\}$. The set of N' bandpass modulator circuitry $213-\{1, \dots, N_b\}$ can be configured as I/Q DAC circuitry or I/Q modulator circuitry, can generate N' active bandpass components $214-\{1, \dots, N'\}$. The set of N' bandpass modulator circuitry $213-\{1, \dots, N_b\}$ can generate the set of bandpass components $214-\{1, \dots, N'\}$ having the carrier frequency f_c . The secure mapper circuitry 211 can also generate control information to activate or deactivate the mixer circuitry $215-\{O_1, \dots, O_{N_b}\}$.

In another embodiment, the secure mapper circuitry 211 can provide control information to the set of active bandpass modulator circuitry $213-\{1, \dots, N_b\}$ to control the initial phase of the generated bandpass components $214-\{1, \dots, N'\}$.

In another embodiment, the set of bandpass modulator circuitry $213-\{1, \dots, N_b\}$ can generate a set of N' active bandpass components $214-\{1, \dots, N'\}$ having the intermediate frequency f_i and the mixer circuitry $215-\{O_1, \dots, O_{N_b}\}$ can upconvert the bandpass components $214-$

$\{1, \dots, N\}$ to the carrier frequency f_c to obtain the carrier frequency components $\mathbf{216}\text{-}\{1, \dots, N_b\}$.

In FIG. 2A, the set of mixer circuitry **215**- $\{O_1, \dots, O_{N_b}\}$ can receive signal **201s-o1** having a certain reference frequency.

In another embodiment, the set of mixer circuitry **215**- $\{O_1, \dots, O_{N_b}\}$ can receive signal **201**-ofc having the carrier frequency f_c .

In another embodiment, the certain reference signal is associated with the intermediate frequency f_i .

In another embodiment, the certain reference signal is associated with up conversion from the intermediate frequency f_i to the carrier frequency f_c .

In another embodiment, the bandpass modulator circuitry **213**- $\{1, \dots, N_b\}$ can output feedback signals $F\{-O_1, \dots, O_{N_b}\}$ to the secure mapper circuitry **211** to generate time or phase control information to enable time or phase synchronization of the active bandpass components **214**- $\{1, \dots, N'\}$.

In another embodiment, the mixer circuitry **215**- $\{O_1, \dots, O_{N_b}\}$ can output the feedback signals F - $\{O_1, \dots, O_{N_b}\}$ to the secure mapper circuitry **211** to generate time or phase control information to enable time or phase synchronization of the active bandpass components **214**- $\{1_i, \dots, N'\}$.

In another embodiment, an output circuitry **225** can include a combiner circuitry **217**. The combiner circuitry **217** can combine the components **216**- $\{1, \dots, N_b\}$ to obtain the output signal **218**.

In another exemplary embodiment, the output circuitry 225 can include a set of amplifiers (e.g., power amplifiers) with each amplifier corresponding to a certain component signal 216- $\{1, \dots, N_b\}$. The set of amplifiers can amplify the components 216- $\{1, \dots, N_b\}$ to obtain a set of amplified signals. The combiner circuitry 225 can then combine the set of amplified signals to obtain the output signal 218.

In another embodiment, the sampling circuitry **221**, the mapper circuitry **208**, the secure mapper circuitry **211**, and the quantizer circuitry **203** can be configured in a single circuitry having hardware, software or firmware, including digital or analog circuitry with or without a microprocessor, a field programmable gate array (FPGA), or a digital signal processor (DSP).

In another embodiment, the mapper circuitry **208** and the secure mapper circuitry **211** can be configured in a single circuitry **223** having hardware, software or firmware, including digital or analog circuitry with or without a microprocessor, an FPGA, a DSP, or the like.

FIG. 2B illustrates one embodiment of a system **200b** of physical layer security in accordance with various aspects as described herein. In FIG. 2B, the system **200b** implements the method **100** of FIG. 1 with N_b quantization bits and $N_b = N_b + 1$ possible bandpass components **214**- $\{1, \dots, N_b + 1\}$. The clock signal **201-b** can be a clock reference signal having the sampling frequency f_s and can be input to the sampling circuitry **221**, the quantizer circuitry **203**, the mapper circuitry **208**, the secure mapper circuitry **211**, the set of bandpass modulator circuitry **213**- $\{1, \dots, N_b + 1\}$, the combiner circuitry **217**, the like, or any combination thereof.

In FIG. 2B, the mapper circuitry **208** generates a set of $N' \leq N_b$ active pairs of in-phase and quadrature-phase components related to $N' \leq N_b$, mapping components according to a mapping table, stored in a memory element (e.g., LUT) configured to include the mapping codes and mapping coefficients associated with the bandpass components, and may also include the quantization bit combinations, the sets of discrete amplitudes a_i , and the set of quantized values. In

this exemplary embodiment, the system **200b** is configured for $N_b=4$ quantization bits and $N_b=N_b+1$ possible bandpass components. Other embodiments can be implemented employing an alternative number of quantization bits. The mapper circuitry **208** can include quantization bits and a quantization encoding table to define a mapping table with mapping codes. The mapper circuitry **208** can include the quantization table to define the set of mapping codes $M_i=\{M_0; M_1; \dots; M_{N_c}\}$, with $N_c=2^{N_b}-1$, which are associated with the sets of mapping coefficients $C_{N_b}^{(0)} \dots C_1^{(0)}$, $C_{N_b}^{(N_c)} \dots C_1^{(N_c)}$ and defined in terms of mapping coefficients $C_k=\{C_1; \dots; C_k\}$, with $k=1, \dots, N_b$. For example, with $N_b=4$ quantization bits and $N_b=5$ results in $N_c=16$ mapping codes $M_i=\{M_0; \dots; M_{15}\}$ directly related to the sets of mapping coefficients $C_5^{(0)}C_4^{(0)}C_3^{(0)}C_2^{(0)}C_1^{(0)}$, $C_5^{(15)}C_4^{(15)}C_3^{(15)}C_2^{(15)}C_1^{(15)}$. The coefficients C_k , $k=1, \dots, N_b+1$, when security is not applied, can map the quantized magnitude value into $N' \leq N_b$ pairs of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ mapping components. C_1 mapping coefficients can represent values from a set of amplitudes $C_1^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_1 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 or the amplitude quantization error Δ . C_2 mapping coefficients can represent values from a set of amplitudes $C_2^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_2 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , or the amplitude quantization error Δ . C_3 mapping coefficients can represent values from a set of amplitudes $C_3^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_3 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , or the amplitude quantization error Δ . C_4 mapping coefficients can represent values from a set of amplitudes $C_4^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_4 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , or the amplitude quantization error Δ . C_5 mapping coefficients can represent values from a set of amplitudes $C_5^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_5 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , the discrete amplitudes of the fourth quantization component a_4 , or the amplitude quantization error Δ .

In another embodiment, when security is not applied, the coefficients C_k , $k=1, \dots, N_b+1$ can map the quantized magnitude value into $N \leq N_b$ pairs of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ mapping components. C_1 mapping coefficients can represent values from a set of amplitudes $C_1^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_1 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 or the amplitude quantization error Δ . C_2 mapping coefficients can represent values from a set of amplitudes $C_2^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_2 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , or the amplitude quantization error Δ . C_3 mapping coefficients can represent values from a set of amplitudes $C_3^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_3 with a null

value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , or the amplitude quantization error Δ . C_4 mapping coefficients can represent values from a set of amplitudes $C_4^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_4 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , the discrete amplitudes of the fourth quantization component a_4 , or the amplitude quantization error Δ . C_5 mapping coefficients can represent values from a set of amplitudes $C_5^{(i)}$, $i=0, \dots, 15$, belonging to a discrete alphabet A_5 with a null value and other values defined as a function of the discrete amplitudes of the first quantization component a_1 , the discrete amplitudes of the second quantization component a_2 , the discrete amplitudes of the third quantization component a_3 , the discrete amplitudes of the fourth quantization component a_4 , the discrete amplitudes of the fifth quantization component a_5 , or the amplitude quantization error Δ .

In FIG. 2B, the secure mapper circuitry 211 can receive from the mapper circuitry 208 the mapping codes 209- $\{1, \dots, N_c\}$ and the mapping coefficients 210- $\{1, \dots, N_b+1\}$ and can select the security scheme to apply to the new secure mapping codes and corresponding mapping coefficients. The secure mapper circuitry 211 can apply the security scheme to compute the secure mapping coefficients C_k' , where $k=1, \dots, N_b$, where $N_b=N_b+1$, and can generate the secure mapping tables with new mapping coefficients C_k' needed to achieve security on mapping the quantized value into a maximum number $N' \leq N_b$, among the N_b possible values of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ mapping components 212- $\{1_p, \dots, N_{b'q}\}$, 212- $\{1_q, \dots, N_{b'q}\}$, related to $N' \leq N_b$ active mapping components given by $C_k' \cos(\alpha_n)$ and $C_k' \sin(\alpha_n)$.

In one embodiment, a security scheme can include permutations between mapping codes, phase rotations of mapping coefficients, puncturing of the mapping codes and the mapping coefficients, or scale factor multiplication of the mapping coefficients. The secure mapper circuitry can generate a secure mapping table having the secure mapping codes and corresponding mapping coefficients configured to map the quantized values into bandpass components and can generate numerical sequences with the positions of samples in which secure mapping tables are applied.

In another embodiment, with permutations of mapping codes, a permutation array $\{P_0, P_1, \dots, P_{N_c}\}$, being the values adopted to define a set of permuted mapping codes $M_i'=\{M_{P_0}; M_{P_1}; \dots; M_{P_{N_c}}\}$, in which mapping codes are associated with the corresponding sets of secure mapping coefficients $C_{N_b'}^{(P_i)}, C_{N_b'-1}^{(P_i)}$, with $i=0, \dots, N_c$. For instance, with $N_b=4$ quantization bits with permutations in sixteen (16) mapping codes for a permutation array $P=\{P_0, P_1, \dots, P_{15}\}$, results in a new permuted set of mapping codes $M_i'=\{M_{P_0}; M_{P_1}; \dots; M_{P_{15}}\}$, which are associated with the corresponding sets of mapping coefficients given by: $C_5^{(P_0)}C_4^{(P_0)}C_3^{(P_0)}C_2^{(P_0)}C_1^{(P_0)}$, $C_5^{(P_1)}C_4^{(P_1)}C_3^{(P_1)}C_2^{(P_1)}C_1^{(P_1)}$, $C_5^{(P_2)}C_4^{(P_2)}C_3^{(P_2)}C_2^{(P_2)}C_1^{(P_2)}$, $C_5^{(P_3)}C_4^{(P_3)}C_3^{(P_3)}C_2^{(P_3)}C_1^{(P_3)}$, $C_5^{(P_4)}C_4^{(P_4)}C_3^{(P_4)}C_2^{(P_4)}C_1^{(P_4)}$, $C_5^{(P_5)}C_4^{(P_5)}C_3^{(P_5)}C_2^{(P_5)}C_1^{(P_5)}$, $C_5^{(P_6)}C_4^{(P_6)}C_3^{(P_6)}C_2^{(P_6)}C_1^{(P_6)}$, $C_5^{(P_7)}C_4^{(P_7)}C_3^{(P_7)}C_2^{(P_7)}C_1^{(P_7)}$, $C_5^{(P_8)}C_4^{(P_8)}C_3^{(P_8)}C_2^{(P_8)}C_1^{(P_8)}$, $C_5^{(P_9)}C_4^{(P_9)}C_3^{(P_9)}C_2^{(P_9)}C_1^{(P_9)}$, $C_5^{(P_{10})}C_4^{(P_{10})}C_3^{(P_{10})}C_2^{(P_{10})}C_1^{(P_{10})}$, $C_5^{(P_{11})}C_4^{(P_{11})}C_3^{(P_{11})}C_2^{(P_{11})}C_1^{(P_{11})}$, $C_5^{(P_{12})}C_4^{(P_{12})}C_3^{(P_{12})}C_2^{(P_{12})}C_1^{(P_{12})}$, $C_5^{(P_{13})}C_4^{(P_{13})}C_3^{(P_{13})}C_2^{(P_{13})}C_1^{(P_{13})}$, $C_5^{(P_{14})}C_4^{(P_{14})}C_3^{(P_{14})}C_2^{(P_{14})}C_1^{(P_{14})}$, $C_5^{(P_{15})}C_4^{(P_{15})}C_3^{(P_{15})}C_2^{(P_{15})}C_1^{(P_{15})}$. For instance, the permutation array $P=\{7, 0, 10, 12, 6, 1, 11, 13, 4, 2, 9, 15, 5, 3, 8, 14\}$, results in the secure mapping table obtained with permutations of mapping codes as shown in Table III below.

TABLE III

Example of Mapping Table After Permutation of Mapping Codes						
C_5	C_4	C_3	C_2	C_1	Mapping Code M_i'	
$C_5^{(p0)}$	$C_4^{(p0)}$	$C_3^{(p0)}$	$C_2^{(p0)}$	$C_1^{(p0)}$	$M'_0 =$	M_{P_0}
$C_5^{(p1)}$	$C_4^{(p1)}$	$C_3^{(p1)}$	$C_2^{(p1)}$	$C_1^{(p1)}$	$M'_1 =$	M_{P_1}
$C_5^{(p2)}$	$C_4^{(p2)}$	$C_3^{(p2)}$	$C_2^{(p2)}$	$C_1^{(p2)}$	$M'_2 =$	M_{P_2}
$C_5^{(p3)}$	$C_4^{(p3)}$	$C_3^{(p3)}$	$C_2^{(p3)}$	$C_1^{(p3)}$	$M'_3 =$	M_{P_3}
$C_5^{(p4)}$	$C_4^{(p4)}$	$C_3^{(p4)}$	$C_2^{(p4)}$	$C_1^{(p4)}$	$M'_4 =$	M_{P_4}
$C_5^{(p5)}$	$C_4^{(p5)}$	$C_3^{(p5)}$	$C_2^{(p5)}$	$C_1^{(p5)}$	$M'_5 =$	M_{P_5}
$C_5^{(p6)}$	$C_4^{(p6)}$	$C_3^{(p6)}$	$C_2^{(p6)}$	$C_1^{(p6)}$	$M'_6 =$	M_{P_6}
$C_5^{(p7)}$	$C_4^{(p7)}$	$C_3^{(p7)}$	$C_2^{(p7)}$	$C_1^{(p7)}$	$M'_7 =$	M_{P_7}
$C_5^{(p8)}$	$C_4^{(p8)}$	$C_3^{(p8)}$	$C_2^{(p8)}$	$C_1^{(p8)}$	$M'_8 =$	M_{P_8}
$C_5^{(p9)}$	$C_4^{(p9)}$	$C_3^{(p9)}$	$C_2^{(p9)}$	$C_1^{(p9)}$	$M'_9 =$	M_{P_9}
$C_5^{(p10)}$	$C_4^{(p10)}$	$C_3^{(p10)}$	$C_2^{(p10)}$	$C_1^{(p10)}$	$M'_{10} =$	$M_{P_{10}}$
$C_5^{(p11)}$	$C_4^{(p11)}$	$C_3^{(p11)}$	$C_2^{(p11)}$	$C_1^{(p11)}$	$M'_{11} =$	$M_{P_{11}}$
$C_5^{(p12)}$	$C_4^{(p12)}$	$C_3^{(p12)}$	$C_2^{(p12)}$	$C_1^{(p12)}$	$M'_{12} =$	$M_{P_{12}}$
$C_5^{(p13)}$	$C_4^{(p13)}$	$C_3^{(p13)}$	$C_2^{(p13)}$	$C_1^{(p13)}$	$M'_{13} =$	$M_{P_{13}}$
$C_5^{(p14)}$	$C_4^{(p14)}$	$C_3^{(p14)}$	$C_2^{(p14)}$	$C_1^{(p14)}$	$M'_{14} =$	$M_{P_{14}}$
$C_5^{(p15)}$	$C_4^{(p15)}$	$C_3^{(p15)}$	$C_2^{(p15)}$	$C_1^{(p15)}$	$M'_{15} =$	$M_{P_{15}}$

The secure mapping table shown in Table III is solely for illustrative purposes and not to be limiting. The number of possible permutation arrays for sixteen (16) mapping codes is $2.0923E+13$, and the permutations of mapping codes can change dynamically over each data block or within a data block according to a predetermined pattern.

In another embodiment, the secure mapper circuitry can apply a puncturing pattern to puncture mapping codes. The secure mapper circuitry can apply a puncturing array $Pu=\{pu_1, \dots, pu_{N_c}\}$, where each pu_i can represent a value of one (1) without puncturing or a value of X for puncturing, with the puncturing pattern to be applied to puncture mapping codes $M_i'=\{M_0; M_1; \dots; M_{N_c}\}$.

In another embodiment, the puncturing rule is constant over the entire data block or may differ for different subsets of samples within the data block. Mapping codes can be punctured according to a predetermined rule and can be replaced by another code or a reference dummy value, and at a receiver a de-puncturing operation can be performed to recover the signal envelope. For instance, a puncturing rule $Pu=\{1, X, 1, 1, X, 1, 1, 1, 1, X, 1, 1, 1, X, 1\}$, results in the punctured mapping codes $M_i'=\{M_0; X; M_2; M_3; X; M_5; \dots; M_9; X; M_{11}; M_{12}; M_{13}; X; M_{15}\}$, directly related to the sets of mapping coefficients: $C_5^{(0)}C_4^{(0)}C_3^{(0)}C_2^{(0)}C_1^{(0)}$, $X, C_5^{(2)}C_4^{(2)}C_3^{(2)}C_2^{(2)}C_1^{(2)}$, $C_5^{(3)}C_4^{(3)}C_3^{(3)}C_2^{(3)}C_1^{(3)}$, $X, C_5^{(5)}C_4^{(5)}C_3^{(5)}C_2^{(5)}C_1^{(5)}$, $\dots$, $C_5^{(9)}C_4^{(9)}C_3^{(9)}C_2^{(9)}C_1^{(9)}$, $X, C_5^{(11)}C_4^{(11)}C_3^{(11)}C_2^{(11)}C_1^{(11)}$, $\dots$, $C_5^{(13)}C_4^{(13)}C_3^{(13)}C_2^{(13)}C_1^{(13)}$, $X, C_5^{(15)}C_4^{(15)}C_3^{(15)}C_2^{(15)}C_1^{(15)}$.

In another embodiment, the puncturing rule can be applied as a random phase rotation $C_k^{(i)}=e^{j\alpha}C_k^{(i)}$ is applied to a subset of mapping codes and corresponding mapping coefficients. The phase rotation can affect all mapping codes or subsets of mapping codes. The phase value can be predetermined and the phase rotation may change over the data block or between successive samples of the input signals' envelope according to a predetermined rule. This corresponds to a phase rotation of the values of the alphabets A_k associated with the sets of possible values of mapping coefficients.

In another embodiment, the phase rotation can be applied to only a set of mapping coefficients $C_{ks}^{(i)}=e^{j\alpha}C_{ks}^{(i)}$, for $i=i_1, \dots, i_n$ and $ks=ks_1, ks_2, \dots, N'$, with $N' < N_b$.

In another embodiment, a scale factor can be applied to a limited set of mapping coefficients $C_{ks}^{(i)}=s_jC_{ks}^{(i)}$, for $i=i_1, \dots, i_n$ and $ks=ks_1, ks_2, \dots, N'$, with $N' < N_b$.

In FIG. 2B, based on a secure mapping table, the secure mapper circuitry 211 can generate a set of $N' \leq N_b$ active

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pairs of in-phase and quadrature-phase mapping components $212-\{1, \dots, N_b\}$, $212-\{1_q, \dots, N_{b'q}\}$ related to $N \leq N_b$, mapping components according to the secure mapping table that can be stored in a memory element (e.g., LUT) that includes the mapping component values according to the secure mapping rule. The secure mapper circuitry **211** can generate numerical sequences with the positions of samples in which the secure mapping tables are applied or numerical sequences are previously stored in the memory elements.

In another embodiment, several sets of numerical sequences having the positions of samples affected by security schemes can be stored in memory elements (e.g., LUTs).

In another embodiment, the mapping table, the secure mapping table, the quantization encoding table, the quantization values table, the quantization amplitude components, or the like can be stored in memory elements (e.g., LUTs) that are operationally coupled to the quantizer circuitry, the mapper circuitry, the secure mapper circuitry, or another circuit.

In another embodiment, the permutation rules, the puncturing rules, the phase rotation rules to be applied in accordance with the secure mapping tables can also be stored in memory elements (e.g., LUTs).

In FIG. 2B, the secure mapper circuitry **211** can output active $N \leq N_b$ pairs of in-phase and quadrature-phase mapping components $212-\{1, \dots, N_b\}$, $212-\{1_q, \dots, N_{b'q}\}$ to a set of N' bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$, which can be configured as I/Q DACs or I/Q modulators, operable to generate the N' active bandpass components $214-\{1, \dots, N_b+1\}$. Without the mixer circuitry $215-\{O_1, \dots, O_{N_b}\}$, the set of N' bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$ can generate the set of N' active bandpass components $214-\{1, \dots, N'\}$ with the carrier frequency f_c .

In another embodiment, the secure mapper circuitry **211** can apply the phase α_n of each sample to control the initial phase of the bandpass components $214-\{1, \dots, N_b+1\}$ generated by the set of bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$.

In another embodiment, the set of bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$ can generate the set of active bandpass components $214-\{1, \dots, N_b+1\}$ at intermediate frequency f_i and the mixer circuitry $215-\{O_1, \dots, O_{N_b}\}$ can be applied to upconvert the bandpass components $214-\{1, \dots, N_b+1\}$ to the carrier frequency f_c .

In another embodiment, the feedback signals $219-\{O_1, \dots, O_{N_b+1}\}$ outputted by the set of bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$ can be input to the secure mapper circuitry **211** operable to generate phase and time control information for the bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$ to assure time and phase synchronization of active bandpass components.

In another embodiment, the feedback signals $219-\{O_1, \dots, O_{N_b+1}\}$ can be output from the set of mixer circuitry $215-\{O_1, \dots, O_{N_b+1}\}$ and input to the secure mapper circuitry **211** to generate phase and time control information to enable time and phase synchronization of active bandpass components.

In another embodiment, the reference signal clock $201s-o1$ having the intermediate frequency f_i can be input to the set of bandpass modulator circuitry $213-\{1, \dots, N_b+1\}$ and the input signal $201-ofc$ having the carrier frequency f_c can provide a reference signal to the set of mixer circuitry $215-\{O_1, \dots, O_{N_b+1}\}$ to enable up-conver-

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sion of the bandpass components $214-\{1, \dots, N_b+1\}$ to obtain carrier frequency components $216-\{1, \dots, N_b+1\}$.

In another exemplary embodiment, the output circuitry **225** can include the combiner circuitry **217**. The combiner circuitry **217** can combine the components $216-\{1, \dots, N_b+1\}$ to obtain the output signal **218**.

In another exemplary embodiment, the output circuitry **225** can include a set of amplifiers (e.g., power amplifiers) with each amplifier corresponding to a certain component signal $216-\{1, \dots, N_b+1\}$. The set of amplifiers can amplify the components $216-\{1, \dots, N_b+1\}$ to obtain a set of amplified signals. The combiner circuitry **225** can then combine the set of amplified signals to obtain the output signal **218**.

FIG. 3 illustrates another embodiment of a method **300** of providing physical layer security in accordance with various aspects as described herein. In FIG. 3, the method **300** may start, for instance, at block **301** where it can include receiving samples of an input signal having information. At block **303**, the method **300** can include quantizing an amplitude of the input signal sample to obtain a set of quantization bits that represents the quantized amplitude of the input signal sample. At block **305**, the method **300** can include determining an amplitude quantization error of the input signal sample based on the difference between the amplitude of the input signal sample and the quantized amplitude of the input signal sample. At block **307**, the method **300** can include determining a compensation factor based on the amplitude quantization error. At block **309**, the method **300** can include determining a set of mapping codes or a set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, the compensation factor of the input signal sample, or a set of discrete amplitudes that corresponds to the set of quantization bits. At block **311**, the method **300** can include selecting a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients. At block **313**, the method **300** can include applying the selected security scheme to the set of mapping codes or the sets of mapping coefficients. At block **315**, the method **300** can include mapping the set of quantization bits that represents the quantized amplitude, the phase or the compensation factor of the input signal sample to obtain a set of mapping components (e.g., in-phase and quadrature-phase mapping components) based on the set of mapping codes or the corresponding sets of mapping coefficients. At block **317**, the method **300** can include modulating the secured set of mapping components to obtain a secured set of bandpass components. At block **319**, the method **300** can include upconverting the secured set of bandpass components to an intermediate frequency or a carrier frequency to obtain a secured set of bandpass components at the Intermediate Frequency (IF) or Radio Frequency (RF). At block **321**, the method **300** can include combining the secured set of bandpass components at the IF or RF frequency to obtain a secured output signal. At block **323**, the method **300** includes outputting the secured output signal that represents the secured set of bandpass components.

Although the subject matter described herein may be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein may also be described in relation to a wired or wireless network, such as the example wireless and wired networks illustrated in FIG. 4. For simplicity, the wired network portion of FIG. 4 depicts network **406** communicatively coupled to network nodes **460**, **460b**, and the wireless network portion of FIG.

4 depicts wireless devices **410**, **410b** and **410c** communicatively coupled to network nodes **460**, **460b**. In practice, a wired and wireless network may further include any additional elements suitable to support communication between wired and wired devices, wireless and wireless devices, 5 wired and wireless devices, or between a wired or wireless device and another communication device, such as a land-line telephone, a service provider, or any other network node or end device. Of the illustrated components, network node **460** and wireless device **410** are depicted with additional detail. Network **406** may provide communication and other 10 types of services to one or more devices to facilitate the devices' access to and/or use of the services provided by, or via, the network **406**.

Network **406** may comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, network **406** may be configured to operate according to specific standards or other types of predefined rules or procedures. Thus, particular embodiments of network **406** may implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), Narrowband Internet of Things (NB-IoT), and/or other suitable 2G, 3G, 4G, 5G, 6G standards; 15 wireless local area network (WLAN) standards, such as the IEEE 402.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

Network **406** may comprise one or more backhaul networks, core networks, internet protocol (IP) networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication 20 between devices.

Network node **460** and wireless device **410** comprise various components described in more detail below. These components work together in order to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network may comprise any number of wired or wireless networks, network nodes, base stations, 25 controllers, wireless devices, relay stations, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

As used herein, network node refers to equipment capable, configured, arranged and/or operable to communicate directly or indirectly with a wired or wireless device and/or with other network nodes or equipment in the network to enable and/or provide access to the device and/or to perform other functions (e.g., administration) in the network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs), and NR NodeBs (gNBs)). Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and may then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also 50 include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio

units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS). Yet further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node may be a virtual network node as described in more detail below. More generally, however, network nodes may represent any suitable wired or wireless device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wired or wireless device with access to the network or to provide some service to a wired or wireless device that has accessed the network.

In FIG. 4, network node **460** includes processing circuitry **470**, device readable medium **480**, interface **490**, auxiliary equipment **454**, power source **486**, power circuitry **487**, antenna **462**, or the like. Although network node **460** illustrated in the example wireless network of FIG. 4 may represent a device that includes the illustrated combination of hardware components, other embodiments may comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Moreover, while the components of network node **460** are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, a network node may comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium **480** may comprise multiple separate hard drives as well as multiple random access memory (RAM) modules). 30

Similarly, network node **460** may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which network node **460** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, network node **460** may be configured to support multiple radio access technologies (RATs). In such embodiments, some components may be duplicated (e.g., separate device readable medium **480** for the different RATs) and some components may be reused (e.g., the same antenna **462** may be shared by the RATs). Network node **460** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **460**, such as, for example, GSM, code division access (CDMA), wideband CDMA (WCDMA), LTE, NR, Wi-Fi, or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **460**. 55

Processing circuitry **470** is configured to perform any determining, calculating, or similar operations (e.g., certain

obtaining operations) described herein as being provided by a network node. These operations performed by processing circuitry 470 may include processing information obtained by processing circuitry 470 by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

Processing circuitry 470 may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuitry, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node 460 components, such as device readable medium 480, network node 460 functionality. For example, processing circuitry 470 may execute instructions stored in device readable medium 480 or in memory within processing circuitry 470. Such functionality may include providing any of the various wireless features, functions, or benefits discussed herein. In some embodiments, processing circuitry 470 may include a system on a chip (SOC).

In some embodiments, processing circuitry 470 may include one or more of RF transceiver circuitry 472 and baseband circuitry 474. In some embodiments, RF transceiver circuitry 472 and baseband circuitry 474 may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry 472 and baseband circuitry 474 may be on the same chip or set of chips, boards, or units. The embodiments described by this disclosure can be implemented for the network node 460 in the interface 490 and/or the processing circuitry 470. Further, the embodiments described by this disclosure can be implemented for the wireless device or user equipment 410 in the interface 414 and/or the processing circuitry 420.

In certain embodiments, some or all the functionality described herein as being provided by a network node, base station, eNB or other such network device may be performed by processing circuitry 470 executing instructions stored on device readable medium 480 or memory within processing circuitry 470. In alternative embodiments, some or all of the functionality may be provided by processing circuitry 470 without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry 470 may be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry 470 alone or to other components of network node 460 but are enjoyed by network node 460 as a whole, and/or by end users and the wireless network generally.

Device readable medium 480 may comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, RAM, ROM, mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by processing circuitry 470. Device

readable medium 480 may store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry 470 and, utilized by network node 460. Device readable medium 480 may be used to store any calculations made by processing circuitry 470 and/or any data received via interface 490. In some embodiments, processing circuitry 470 and device readable medium 480 may be considered to be integrated.

Interface 490 is used in the wired or wireless communication of signaling and/or data between network node 460, network 406, and/or wireless devices 410. As illustrated, interface 490 comprises port(s)/terminal(s) 494 to send and receive data, for example to and from network 406 over a wired connection. Interface 490 also includes radio front end circuitry 492 that may be coupled to, or in certain embodiments a part of, antenna 462. Radio front end circuitry 492 comprises filters 498 and power amplifiers 496. Radio front end circuitry 492 may be coupled to antenna 462 and processing circuitry 470. Radio front end circuitry 492 may receive digital data that is to be sent out to other network nodes or wireless devices via a wireless connection. Radio front end circuitry 492 may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters 498 and/or amplifiers 496. The radio signal may then be transmitted via antenna 462. Similarly, when receiving data, antenna 462 may collect radio signals which are then converted into digital data by radio front end circuitry 492. The digital data may be passed to processing circuitry 470. In other embodiments, the interface may comprise different components and/or different combinations of components.

In certain alternative embodiments, network node 460 may not include separate radio front end circuitry 492, instead, processing circuitry 470 may comprise radio front end circuitry and may be coupled to antenna 462 without separate radio front end circuitry 492. Similarly, in some embodiments, all or some of RF transceiver circuitry 472 may be considered a part of interface 490. In still other embodiments, interface 490 may include one or more ports or terminals 494, radio front end circuitry 492, and RF transceiver circuitry 472, as part of a radio unit (not shown), and interface 490 may communicate with baseband circuitry 474, which is part of a digital unit (not shown).

Antenna 462 may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna 462 may be coupled to radio front end circuitry 492 and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna 462 may comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and 6 GHz. A_n omni-directional antenna may be used to transmit/receive radio signals in any direction, a sector antenna may be used to transmit/receive radio signals from devices within a particular area, and a panel antenna may be a line-of-sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna may be referred to as multiple-input multiple-output (MIMO). In certain embodiments, antenna 462 may be separate from network node 460 and may be connectable to network node 460 through an interface or port.

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Antenna **462**, interface **490**, and/or processing circuitry **470** may be configured to perform any receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals may be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna **462**, interface **490**, and/or processing circuitry **470** may be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals may be transmitted to a wireless device, another network node and/or any other network equipment.

Power circuitry **487** may comprise, or be coupled to, power management circuitry and is configured to supply the components of network node **460** with power for performing the functionality described herein. Power circuitry **487** may receive power from power source **486**. Power source **486** and/or power circuitry **487** may be configured to provide power to the various components of network node **460** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source **486** may either be included in, or external to, power circuitry **487** and/or network node **460**. For example, network node **460** may be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry **487**. As a further example, power source **486** may comprise a source of power in the form of a battery or battery pack which is coupled to, or integrated in, power circuitry **487**. The battery may provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, may also be used.

Alternative embodiments of network node **460** may include additional components beyond those shown in FIG. **4** that may be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node **460** may include user interface equipment to allow input of information into network node **460** and to allow output of information from network node **460**. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for network node **460**.

As used herein, wireless device refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other wireless devices. Unless otherwise noted, the term wireless device may be used interchangeably herein with user equipment (UE). Communicating wirelessly may involve transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. In some embodiments, a wireless device may be configured to transmit and/or receive information without direct human interaction. For instance, a wireless device may be designed to transmit information to a network on a predetermined schedule, when triggered by an internal or external event, or in response to requests from the network. Examples of a wireless device include, but are not limited to, a smart phone, a mobile phone, a cell phone, a voice over IP (VOIP) phone, a wireless local loop phone, a desktop computer, a personal digital assistant (PDA), a wireless cameras, a gaming console or device, a music storage device, a play-back appliance, a wearable terminal device, a wireless endpoint, a mobile station, a tablet, a laptop, a laptop-embedded equipment (LEE), a laptop-mounted equipment

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(LME), a smart device, a wireless customer-premise equipment (CPE), a vehicle-mounted wireless terminal device, etc., A wireless device may support device-to-device (D2D) communication, for example by implementing a 3rd Generation Partnership Project (3GPP) standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and may in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a wireless device may represent a machine or other device that performs monitoring and/or measurements, and transmits the results of such monitoring and/or measurements to another wireless device and/or a network node. The wireless device may in this case be a machine-to-machine (M2M) device, which may in a 3GPP context be referred to as an MTC device. As one particular example, the wireless device may be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Particular examples of such machines or devices are sensors, metering devices such as power meters, industrial machinery, or home or personal appliances (e.g. refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a wireless device may represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A wireless device as described above may represent the endpoint of a wireless connection, in which case the device may be referred to as a wireless terminal. Furthermore, a wireless device as described above may be mobile, in which case it may also be referred to as a mobile device or a mobile terminal.

As illustrated, wireless device **410** includes antenna **411**, interface **414**, processing circuitry **420**, device readable medium **430**, user interface equipment **432**, auxiliary equipment **434**, power source **436** power circuitry **437**, or the like. Wireless device **410** may include multiple sets of one or more of the illustrated components for different wireless technologies supported by wireless device **410**, such as, for example, GSM, WCDMA, LTE, NR, Wi-Fi, WiMAX, NB-IoT, or Bluetooth wireless technologies, just to mention a few. These wireless technologies may be integrated into the same or different chips or set of chips as other components within wireless device **410**.

Antenna **411** may include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is coupled to interface **414**. In certain alternative embodiments, antenna **411** may be separate from wireless device **410** and be connectable to wireless device **410** through an interface or port. Antenna **411**, interface **414**, and/or processing circuitry **420** may be configured to perform any receiving or transmitting operations described herein as being performed by a wireless device. Any information, data and/or signals may be received from a network node and/or another wireless device. In some embodiments, radio front end circuitry and/or antenna **411** may be considered an interface.

As illustrated, interface **414** comprises radio front end circuitry **412** and antenna **411**. Radio front end circuitry **412** comprises one or more filters **418** and power amplifiers **416**. Radio front end circuitry **412** is coupled to antenna **411** and processing circuitry **420** and is configured to condition signals communicated between antenna **411** and processing circuitry **420**. Radio front end circuitry **412** may be coupled to or a part of antenna **411**. In some embodiments, wireless device **410** may not include separate radio front end circuitry **412**; rather, processing circuitry **420** may comprise radio front end circuitry and may be coupled to antenna **411**.

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Similarly, in some embodiments, some or all of RF transceiver circuitry 422 may be considered a part of interface 414. Radio front end circuitry 412 may receive digital data that is to be sent out to other network nodes or wireless devices via a wireless connection. Radio front end circuitry 412 may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters 418 and/or amplifiers 416. The radio signal may then be transmitted via antenna 411. Similarly, when receiving data, antenna 411 may collect radio signals which are then converted into digital data by radio front end circuitry 412. The digital data may be passed to processing circuitry 420. In other embodiments, the interface may comprise different components and/or different combinations of components.

Processing circuitry 420 may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuitry, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide, either alone or in conjunction with other wireless device 410 components, such as device readable medium 430, wireless device 410 functionality. Such functionality may include providing any of the various wireless features or benefits discussed herein. For example, processing circuitry 420 may execute instructions stored in device readable medium 430 or in memory within processing circuitry 420 to provide the functionality disclosed herein.

As illustrated, processing circuitry 420 includes one or more of RF transceiver circuitry 422, baseband circuitry 424, and application processing circuitry 426. In other embodiments, the processing circuitry may comprise different components and/or different combinations of components. In certain embodiments processing circuitry 420 of wireless device 410 may comprise a SOC. In some embodiments, RF transceiver circuitry 422, baseband circuitry 424, and application processing circuitry 426 may be on separate chips or sets of chips. In alternative embodiments, part or all of baseband circuitry 424 and application processing circuitry 426 may be combined into one chip or set of chips, and RF transceiver circuitry 422 may be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry 422 and baseband circuitry 424 may be on the same chip or set of chips, and application processing circuitry 426 may be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry 422, baseband circuitry 424, and application processing circuitry 426 may be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry 422 may be a part of interface 414. RF transceiver circuitry 422 may condition RF signals for processing circuitry 420.

In certain embodiments, some or all of the functionality described herein as being performed by a wireless device may be provided by processing circuitry 420 executing instructions stored on device readable medium 430, which in certain embodiments may be a computer-readable storage medium. In alternative embodiments, some or all of the functionality may be provided by processing circuitry 420 without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry 420 may be configured to perform the described functionality. The benefits provided by such functionality are not limited to

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processing circuitry 420 alone or to other components of wireless device 410 but are enjoyed by wireless device 410 as a whole, and/or by end users and the wireless network generally.

Processing circuitry 420 may be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a wireless device. These operations, as performed by processing circuitry 420, may include processing information obtained by processing circuitry 420 by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored by wireless device 410, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

Device readable medium 430 may be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry 420. Device readable medium 430 may include computer memory (e.g., RAM or ROM), mass storage media (e.g., a hard disk), removable storage media (e.g., a CD or a DVD), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that may be used by processing circuitry 420. In some embodiments, processing circuitry 420 and device readable medium 430 may be considered to be integrated.

User interface equipment 432 may provide components that allow for a human user to interact with wireless device 410. Such interaction may be of many forms, such as visual, aural, tactile, etc. User interface equipment 432 may be operable to produce output to the user and to allow the user to provide input to wireless device 410. The type of interaction may vary depending on the type of user interface equipment 432 installed in wireless device 410. For example, if wireless device 410 is a smart phone, the interaction may be via a touch screen; if wireless device 410 is a smart meter, the interaction may be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment 432 may include input interfaces, devices and circuitry, and output interfaces, devices and circuitry. User interface equipment 432 is configured to allow input of information into wireless device 410 and is coupled to processing circuitry 420 to allow processing circuitry 420 to process the input information. User interface equipment 432 may include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment 432 is also configured to allow output of information from wireless device 410, and to allow processing circuitry 420 to output information from wireless device 410. User interface equipment 432 may include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices and circuitry of user interface equipment 432, wireless device 410 may communicate with end users and/or the wireless network and allow them to benefit from the functionality described herein.

Auxiliary equipment 434 is operable to provide more specific functionality which may not be generally performed by wireless devices. This may comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired commu-

nications etc. The inclusion and type of components of auxiliary equipment **434** may vary depending on the embodiment and/or scenario.

Power source **436** may, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, may also be used. The wireless device **410** may further comprise power circuitry **437** for delivering power from power source **436** to the various parts of wireless device **410** which need power from power source **436** to carry out any functionality described or indicated herein. Power circuitry **437** may in certain embodiments comprise power management circuitry. Power circuitry **437** may additionally or alternatively be operable to receive power from an external power source; in which case wireless device **410** may be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry **437** may also be operable in certain embodiments to deliver power from an external power source to power source **436**. This may be, for example, for the charging of power source **436**. Power circuitry **437** may perform any formatting, converting, or other modification to the power from power source **436** to make the power suitable for the respective components of wireless device **410** to which power is supplied.

FIG. 5 illustrates one embodiment of an electronic device **500** in accordance with various aspects described herein. As used herein, the device **500** can be a wired device or a wireless device. Further, the device **500** may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, the device **500** may represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller, IoT device). Alternatively, the device **500** may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter). The device **500** may be any device identified by the 3rd 3GPP, including a NB-IoT device, a machine type communication (MTC) device, and/or an enhanced MTC (eMTC) device. The device **500**, as illustrated in FIG. 5, is one example of a wired or wireless device such as configured for communication in accordance with one or more communication standards promulgated by the 3GPP, such as 3GPP's GSM, UMTS, LTE, 5G, 6G, Wi-Fi 6 (802.11ax), 3GPP Rel 17, DVB-S2X, wired communications, and satellite communications standards. Accordingly, the components discussed herein can be applicable to a wired or wireless device.

As used herein, a wired device refers to a device that can be communicatively coupled to another device or network via a physical connection. A wired device can include a computer peripheral device, an audio or visual device, a networking device, a game console or accessory device, an industrial or medical equipment device, a home appliance device, an office device, another device, the like, or any combination thereof. A computer peripheral device can include a keyboard (e.g., coupled via USB or PS/2 cable), a mouse (e.g., coupled via USB or PS/2 cable), a monitor (e.g., coupled via HDMI, VGA, DVI, Display Port or USB cable), a printer (e.g., coupled via USB, Ethernet, or parallel port cable), a hard drive (e.g., coupled via USB, Thunderbolt or eSATA cable), a scanner (e.g., coupled via USB or Ethernet cable), a webcam (e.g., coupled via USB cable), the like, or any combination thereof. An audio or visual device

can include a headphone (e.g., coupled via 3.5 mm audio jacks or USB cable), a speaker (e.g., coupled via 3.5 mm jacks, RCA cable or optical audio cable), a microphone (e.g., coupled via XLR, USB, or 3.5 mm cable), a home theater device (e.g., coupled via HDMI, RCA or optical cable), a projector (e.g., coupled via HDMI, VGA, Display Port or USB cable), the like, or any combination thereof. A networking device can include a router (e.g., coupled via Ethernet cable), a modem (e.g., coupled via Ethernet cable), a switch (e.g., coupled via Ethernet cable), an Ethernet adapter (e.g., coupled via Ethernet cable), the like, or any combination thereof. A game console or accessory device can include a game controller (e.g., coupled via USB cables), a gaming headset (e.g., coupled over USB or 3.5 mm jack cable), a console-to-TV connection (e.g., coupled over an HDMI or component cable), the like, or any combination thereof. An industrial or medical equipment device can include a CNC machine (e.g., coupled via serial or USB cable), a medical monitor (e.g., ECG, EEG), a laboratory instrument (e.g., coupled via USB or RS232 cable), the like, or any combination thereof. A home appliance device can include a wired security camera (e.g., coupled over coaxial cables), a smart thermostat (e.g., coupled over Ethernet cable), a wired smoke detector (e.g., coupled over a home's electrical system), the like, or any combination thereof. An office equipment device can include a fax machine (e.g., coupled via phone or USB cable), a wired telephone (e.g., coupled via telephone cable), a photocopier (e.g., coupled via USB or Ethernet cable), the like, or any combination thereof. Other devices can include a barcode scanner (e.g., coupled via USB or serial cable), a POS terminal (e.g., coupled via USB or serial cable), a digital camera (e.g., coupled via USB or HDMI cable), a power supply (e.g., coupled via power cable).

In FIG. 5, the device **500** includes processing circuitry **501** that is operatively coupled to input/output interface **505**, RF interface **509**, network connection interface **511**, memory **515** including RAM **517**, ROM **519**, and storage medium **521** or the like, communication subsystem **531**, power source **513**, and/or any other component, or any combination thereof. Storage medium **521** includes operating system **523**, application program **525**, and data **527**. In other embodiments, storage medium **521** may include other similar types of information. Certain devices may utilize all of the components shown in FIG. 5, or only a subset of the components. The level of integration between the components may vary from one device to another device. Further, certain devices may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

In FIG. 5, processing circuitry **501** may be configured to process computer instructions and data. Processing circuitry **501** may be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, application specific integrated circuits (ASICs), etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or a DSP, together with appropriate software; or any combination of the above. For example, the processing circuitry **501** may include two central processing units (CPUs). Data may be information in a form suitable for use by a computer.

In the depicted embodiment, input/output interface **505** may be configured to provide a communication interface to an input device, output device, or input and output device.

device **500** may be configured to use an output device via input/output interface **505**. An output device may use the same type of interface port as an input device. For example, a USB port may be used to provide input to and output from device **500**. The output device may be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. device **500** may be configured to use an input device via input/output interface **505** to allow a user to capture information into device **500**. The input device may include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a user. A sensor may be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device may be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.

In FIG. **5**, RF interface **509** may be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface **511** may be configured to provide a communication interface to network **543a**. Network **543a** may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **543a** may comprise a Wi-Fi network. Network connection interface **511** may be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface **511** may implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions may share circuitry components, software or firmware, or alternatively may be implemented separately.

RAM **517** may be configured to interface via bus **503** to processing circuitry **501** to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM **519** may be configured to provide computer instructions or data to processing circuitry **501**. For example, ROM **519** may be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-volatile memory. Storage medium **521** may be configured to include memory such as RAM, ROM, PROM, EPROM, EEPROM, magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives. In one example, storage medium **521** may be configured to include operating system **523**, application program **525** such as a web browser application, a widget or gadget engine or another application, and data file **527**. Storage medium **521** may store, for use by device **500**, any of a variety of various operating systems or combinations of operating systems.

Storage medium **521** may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory,

USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium **521** may allow device **500** to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system may be tangibly embodied in storage medium **521**, which may comprise a device readable medium.

In FIG. **5**, processing circuitry **501** may be configured to communicate with network **543b** using communication subsystem **531**. Network **543a** and network **543b** may be the same network or networks or different network or networks. Communication subsystem **531** may be configured to include one or more transceivers used to communicate with network **543b**. For example, communication subsystem **531** may be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another wireless device, device, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.11, CDMA, WCDMA, GSM, LTE, UMTS terrestrial radio access network (UTRAN), WiMax, or the like. Each transceiver may include transmitter **533** and/or receiver **535** to implement transmitter or receiver functionality, respectively, appropriate to the RAN links (e.g., frequency allocations and the like). Further, transmitter **533** and receiver **535** of each transceiver may share circuitry components, software or firmware, or alternatively may be implemented separately.

In the illustrated embodiment, the communication functions of communication subsystem **531** may include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem **531** may include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network **543b** may encompass wired and/or wireless networks such as a LAN, a WAN, a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **543b** may be a cellular network, a Wi-Fi network, and/or a near-field network. Power source **513** may be configured to provide alternating current (AC) or direct current (DC) power to components of device **500**.

The features, benefits and/or functions described herein may be implemented in one of the components of device **500** or partitioned across multiple components of device **500**. Further, the features, benefits, and/or functions described herein may be implemented in any combination of hardware, software or firmware. In one example, communication subsystem **531** may be configured to include any of the components described herein. Further, processing circuitry **501** may be configured to communicate with any of such components over bus **503**. In another example, any of such components may be represented by program instructions stored in memory that when executed by processing cir-

cuitry **501** perform the corresponding functions described herein. In another example, the functionality of any of such components may be partitioned between processing circuitry **501** and communication subsystem **531**. In another example, the non-computationally intensive functions of any of such components may be implemented in software or firmware and the computationally intensive functions may be implemented in hardware. The embodiments described by this disclosure can be implemented for the device **500** in the RF interface circuitry **509** or the transmitter circuitry **533** of the communication subsystem **531**.

Those skilled in the art will also appreciate that embodiments herein further include corresponding computer programs.

A computer program comprises instructions which, when executed on at least one processor of an apparatus, cause the apparatus to carry out any of the respective processing described above. A computer program in this regard may comprise one or more code modules corresponding to the means or units described above.

Embodiments further include a carrier containing such a computer program. This carrier may comprise one of an electronic signal, optical signal, radio signal, or computer readable storage medium.

In this regard, embodiments herein also include a computer program product stored on a non-transitory computer readable (storage or recording) medium and comprising instructions that, when executed by a processor of an apparatus, cause the apparatus to perform as described above.

Embodiments further include a computer program product comprising program code portions for performing the steps of any of the embodiments herein when the computer program product is executed by a computing device. This computer program product may be stored on a computer readable recording medium.

Additional embodiments will now be described. At least some of these embodiments may be described as applicable in certain contexts for illustrative purposes, but the embodiments are similarly applicable in other contexts not explicitly described.

In one exemplary embodiment, a method of physical layer security for codified amplification includes receiving input signal samples s_n of a baseband signal $s_n = s(t_n) = s_{I,n} + js_{Q,n}$, where $s_{I,n}$ and $s_{Q,n}$ represent the respective in-phase and quadrature-phase component of each sample s_n , and t_n is the sampling instant. The method includes receiving clock reference signals associated with a carrier frequency f_c , a sampling rate f_s , or an intermediate frequency f_i . The method further includes determining, for each sample s_n , an amplitude

$$A_n = |s_n| = \sqrt{s_{I,n}^2 + s_{Q,n}^2},$$

and a phase

$$\alpha_n = \tan^{-1}\left(\frac{s_{Q,n}}{s_{I,n}}\right)$$

of that sample s_n . The method further includes quantizing the amplitude A_n as $A_{qn} = A_n + e_q$, where e_q denotes the quantization error of that sample s_n based on a quantization table with N_b bits, a table with the finite set of quantization values, and a table with the discrete amplitudes of quantization

components $a_i = \{a_1, a_2, \dots, a_{N_b}\}$, with $N_b = N_b$ or $N_b = N_b + 1$, in which the quantized value is decomposed, and based on an amplitude quantization error e_q computing a compensation factor Δ belonging to a finite discrete alphabet. The method further includes determining a mapping table with a set of mapping codes $M_i = \{M_0; \dots; M_{N_c}\}$ with $N_c = 2^{N_b} - 1$, based on the magnitude A_{qn} , the phase α_n , the quantization error compensation factor Δ and the quantization table, which are associated with the sets of mapping coefficients values $C_{N_b}^{(0)} \dots C_1^{(0)}, \dots, C_{N_b}^{(N_c)} \dots C_1^{(N_c)}$, with $N_b = N_b$ or $N_b = N_b + 1$, which are defined in terms of mapping coefficients $C_k = \{C_k^{(1)}; \dots; C_k^{(N_c)}\}$, with $k = 1, \dots, N_b$, with the mapping coefficient values belonging to finite alphabets A_k . The method further includes selecting, based on the coded mapping table and the quantization table, a security scheme based on permutations of mapping codes or phase rotation or puncturing of mapping codes or scale factor multiplication to be applied in generating secure mapping table with secure mapping codes and mapping coefficients. The method further includes performing a secure mapping that applies the secure mapping table with mapping coefficients C_k' where $k = 1, \dots, N_b$, with $N_b = N_b$ or $N_b = N_b + 1$, to determine secure mapping tables with new mapping coefficients C_k' to enable security on mapping the quantized value into a maximum number $N' \leq N_b$ of active pairs of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ components related to $N' \leq N_b$ active mapping components given by $C_k' \cos(\alpha_n)$ and $C_k' \sin(\alpha_n)$. The method further includes mapping the quantized value into N' active mapping components and generating N' active bandpass components.

In another exemplary embodiment, the method can further include a set of LUTs that include the mapping table, the secure mapping table, the security scheme, the quantization table, the sets of discrete amplitudes a_i , or the set of quantization values.

In another exemplary embodiment, the maximum number of possible bandpass components and mapping coefficients C_k' can be N_b , and the coefficients C_k' can represent values from N_b discrete alphabets.

In another exemplary embodiment, the maximum number of possible bandpass components or mapping coefficients C_k' can be $N_b + 1$ for N_b quantization bits and coefficients C_k' can represent values from $N_b + 1$ discrete alphabets.

In another exemplary embodiment, the bandpass modulator circuitry can be operable to receive the secure mapping coefficients to generate the bandpass components according to the secure mapping table.

In another exemplary embodiment, the set of active bandpass components can be generated at a carrier frequency f_c or an intermediate frequency f_i .

In another exemplary embodiment, the method can further include generating numerical sequences having the positions of the samples to which the security schemes are applied.

In another exemplary embodiment, the numerical sequences can be stored in one or more LUTs.

In another exemplary embodiment, the method can further include generating pairs of in-phase and quadrature-phase mapping components and providing active pairs of mapping components to the corresponding bandpass modulator circuitry.

In another exemplary embodiment, the security scheme can be based on permutations of mapping codes and corresponding mapping coefficients.

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In another exemplary embodiment, the security scheme can be based on phase rotations of the mapping coefficients associated with the mapping codes that can affect all or a subset of the mapping codes.

In another exemplary embodiment, the security scheme can be based on a scale factor multiplication of the mapping coefficients associated with the mapping codes that can affect all or a sub-set of mapping codes; or

In another exemplary embodiment, the security scheme can be based on puncturing of the mapping codes and the corresponding mapping coefficients.

In another exemplary embodiment, the security scheme can include permutations, puncturing, phase rotation and scale factor multiplication of mapping codes, the like, or any combination thereof.

In another exemplary embodiment, the security scheme can be applied at the same time or applied separately to different samples or subsets of samples.

In another exemplary embodiment, any of the security schemes can be combined with interleaving techniques affecting the sequence of samples of the input signal.

In another exemplary embodiment, the method can further include determining control information to activate or deactivate bandpass modulator circuitry associated with the mapping components based on the secure mapping table.

In another exemplary embodiment, the method can further include receiving samples s_n of a baseband signal $s(t)$ and generating in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ components of the samples s_n .

In another embodiment, the method can further include receiving a baseband signal and sampling, by a sampling circuitry or in a sample and hold (S/H) circuitry, the baseband signal to obtain samples of the baseband signal.

In another embodiment, the bandpass modulator circuitry can be I/Q modulator circuitry or I/Q DAC circuitry.

In another exemplary embodiment, the method can further include determining the amplitude A_n of the sample, quantizing the amplitude A_n to obtain the quantized amplitude A_{qn} of the sample, generating the mapping table, generating the secure mapping table, generating the control information, and generating the secure mapping of $N' \leq N_b$ active pairs of in-phase and quadrature-phase components can be performed in a single method step.

In one exemplary embodiment, an apparatus configured to perform physical layer security for codified amplification includes an input circuitry operable to receive samples s_n of a baseband signal $s_n = s(t_n) = s_{I,n} + js_{Q,n}$, where $s_{I,n}$ and $s_{Q,n}$ represent the respective in-phase and quadrature-phase components of each sample s_n and t_n is the sampling instant. The input circuitry is also operable to receive clock reference signals associated with a carrier frequency f_c , a sampling rate f_s , or an intermediate frequency f_j . The apparatus further includes a quantizer circuitry operable to, for each sample s_n , determine an amplitude

$$A_n = |s_n| = \sqrt{s_{I,n}^2 + s_{Q,n}^2},$$

and a phase

$$\alpha_n = \tan^{-1}\left(\frac{s_{Q,n}}{s_{I,n}}\right)$$

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of that sample s_n . The quantizer circuitry is further operable to quantizing the amplitude A_n as $A_{qn} = A_n + e_q$, where e_q denotes the quantization error of that sample s_n based on a quantization table with N_b bits, a table with the finite set of quantization values, and a table with the discrete amplitudes of quantization components $a_i = \{a_1, a_2, \dots, a_{N_b}\}$, with $N_b = N_b$ or $N_b = N_b + 1$, in which the quantized value is decomposed, and based on an amplitude quantization error e_q computing a compensation factor Δ belonging to a finite discrete alphabet. The apparatus further includes a mapper circuitry operable to determine a mapping table having a set of mapping codes $M_i = \{M_0; \dots; M_{N_c}\}$ with $N_c = 2^{N_b} - 1$, based on the magnitude A_q , the phase α_n , the quantization error compensation factor Δ and the quantization table, which are associated with the sets of mapping coefficients values $C_{N_b}^{(0)} \dots C_1^{(0)}, \dots, C_{N_b}^{(N_c)} \dots C_1^{(N_c)}$, with $N_b = N_b$ Or $N_b = N_b + 1$, which are defined in terms of mapping coefficients $C_k = \{C_k^{(1)}; \dots; C_k^{(N_c)}\}$, with $k = 1, \dots, N_b$, with the mapping coefficient values belonging to finite alphabets A_k .

The apparatus further includes a secure mapper circuitry operable to receive the mapping table having mapping codes and mapping coefficients and selects the security scheme to apply towards determining secure mapping coefficients C'_k , $k = 1, \dots, N_b$, and to determine secure mapping tables having new mapping coefficients C'_k required to enable security on mapping the quantized value to a maximum number $N' \leq N_b$ of active pairs of in-phase $s_{I,n}$ and quadrature-phase $s_{Q,n}$ components related to $N' \leq N_b$ active mapping components given by $C'_k \cos(\alpha_n)$ and $C'_k \sin(\alpha_n)$. Further, the secure mapper circuitry is operable to determine a set of $N' \leq N_b$ active pairs of in-phase and quadrature-phase mapping components according to the secure mapping table. In addition, the secure mapper circuitry is operable to generate numerical sequences having the positions of the samples to which the security schemes are applied in the generation of the bandpass components. The secure mapping circuitry is also operable to provide $N' \leq N_b$ active pairs of in-phase and quadrature-phase mapping components to a set of N' bandpass modulator circuitry, which can be I/Q modulator circuitry or I/Q DAC circuitry, operable to generate the N' active bandpass components. The apparatus also includes a set of N_b bandpass modulator circuitry with each being operable to receive the active N' pairs of in-phase and quadrature-phase mapping components and generate N' active bandpass components having the carrier frequency f_c .

In one exemplary embodiment, a method is performed by an electronic device. The method includes outputting an output signal that represents a set of pairs of in-phase and quadrature-phase mapping components mapped from a set of mapping codes and corresponding sets of mapping coefficients based on a set of quantization bits that represents a quantized amplitude of a sample of an input signal having information and a phase of the input signal sample. Further, the set of pairs of in-phase and quadrature-phase mapped components collectively represent the input signal sample. Each mapping code corresponds to a certain set of the sets of mapping coefficients with at least one mapping coefficient in each set of mapping coefficients being related to a set of discrete amplitudes that correspond to the set of quantization bits.

In another exemplary embodiment, the method can further include quantizing the amplitude of the input signal sample to obtain the set of quantization bits that represents the quantized amplitude of the input signal sample.

In another exemplary embodiment, the method can further include mapping the set of quantization bits that represents the quantized amplitude of the input signal sample

and the phase of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

In another exemplary embodiment, the method can further include determining the set of mapping codes or the sets of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, or the set of discrete amplitudes that corresponds to the set of quantization bits.

In another exemplary embodiment, the method can further include selecting a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients; or applying the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients.

In another exemplary embodiment, the physical layer security scheme can include applying a scale factor or a phase rotation to the sets of mapping coefficients.

In another exemplary embodiment, the method can further include determining an amplitude quantization error of the input signal sample based on the amplitude of the input signal sample and the quantized amplitude of the input signal sample; determining an amplitude quantization error compensation factor based on the amplitude quantization error; determining the set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization compensation factor of the input signal sample; or wherein the mapping step includes mapping the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input sample and the amplitude quantization error compensation factor of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

In another exemplary embodiment, the method can further include modulating, by a set of modulator circuitry, the set of pairs of in-phase and quadrature-phase mapped components to obtain a set of modulated signals; or combining, by a combiner circuitry, the set of modulated signals to obtain the output signal.

In one exemplary embodiment, an electronic device is operable to output an output signal that represents a set of pairs of in-phase and quadrature-phase mapped components mapped from a set of mapping codes and corresponding sets of mapping coefficients based on a set of quantization bits that represents a quantized amplitude of a sample of an input signal having information and a phase of the input signal sample. Further, the set of pairs of in-phase and quadrature-phase mapped components collectively represent the input signal sample and each mapping code corresponds to a certain set of the sets of mapping coefficients with at least one mapping coefficient in each set of secured mapping coefficients being related to a set of discrete amplitudes that correspond to the set of quantization bits.

In another exemplary embodiment, the electronic device can further include a quantizer circuitry operable to quantize the amplitude of the input signal sample to obtain the set of quantization bits that represents the quantized amplitude of the input signal sample.

In another exemplary embodiment, the electronic device can further include a mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input

signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

In another exemplary embodiment, the electronic device can further include a mapping coefficient determination circuitry operable to determine the set of mapping codes or the sets of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample, or the set of discrete amplitudes that corresponds to the set of quantization bits.

In another exemplary embodiment, the electronic device can further include a physical layer security scheme selection circuitry operable to select a physical layer security scheme to apply to the set of mapping codes or the sets of mapping coefficients; or a physical layer security scheme application circuitry operable to apply the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients.

In another exemplary embodiment, the electronic device can further include a quantization error determination circuitry operable to determine an amplitude quantization error of the input signal sample based on the amplitude of the input signal sample and the quantized amplitude of the input signal sample; a compensation factor determination circuitry operable to determine an amplitude quantization error compensation factor of the input signal sample based on the amplitude quantization error; a mapping coefficient determination circuitry operable to determine the set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization compensation factor of the input signal sample; or a mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample, the phase of the input signal sample and the amplitude quantization error compensation factor of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients.

In one exemplary embodiment, an electronic device includes a quantizer circuitry operable to quantize an amplitude of a sample of an input signal having information to obtain a set of quantization bits that represent the quantized amplitude of the input signal sample; a mapping coefficient determination circuitry operable to determine a set of mapping coefficients based on the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample; a physical layer security scheme selection circuitry operable to select a physical layer security scheme to apply to the set of pairs of in-phase and quadrature-phase mapped components; a physical layer security scheme application circuitry operable to apply the physical layer security scheme to the set of mapping codes or the sets of mapping coefficients; a mapper circuitry operable to map the set of quantization bits that represents the quantized amplitude of the input signal sample and the phase of the input signal sample to obtain the set of pairs of in-phase and quadrature-phase mapped components based on the set of mapping codes and the corresponding sets of mapping coefficients; a set of modulator circuitry operable to modulate the set of pairs of in-phase and quadrature-phase mapped components to obtain a set of

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modulated signals; and a combiner circuitry operable to combine the set of modulated signals to obtain an output signal.

In one exemplary embodiment, a method is performed by an electronic device having a first circuitry electrically coupled to a set of amplifier circuitry operable to collectively amplify an input signal having information. The method includes outputting, by the first circuitry, a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components. Further, the secured set of bandpass components collectively represent a sample of the input signal and correspond to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

In another exemplary embodiment, each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample.

In another exemplary embodiment, at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

In another exemplary embodiment, the method includes applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtaining the secured set of bandpass components based on the secured set of mapping codes.

In another exemplary embodiment, the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

In another exemplary embodiment, the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

In another exemplary embodiment, the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

In another exemplary embodiment, each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and/or a phase of the polar representation of the input signal sample.

In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to collectively amplify an input signal having information and a first circuitry electrically coupled to the set of amplifier circuitry and operable to output a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components. Further, the secured set of bandpass components collectively represents a sample of the input signal and corresponds to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

In another exemplary embodiment, the first circuitry is further operable to apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtain the secured set of bandpass components based on the secured set of mapping codes.

In one exemplary embodiment, a method performed by an electronic device having a set of amplifier circuitry includes

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amplifying, by the set of amplifier circuitry, a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a physical layer security scheme and is associated with a quantized polar representation of the input signal sample.

In another exemplary embodiment, each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

In another exemplary embodiment, at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

In another exemplary embodiment, the method further includes applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtaining the secured set of bandpass components based on the secured set of mapping codes.

In another exemplary embodiment, each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

In one exemplary embodiment, an electronic device includes a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information. Further, the set of output signals corresponds to a set of bandpass components that is secured by a physical layer security scheme and is associated with a quantized polar representation of the input signal sample.

In another exemplary embodiment, the device further includes a first circuitry electrically coupled to the set of amplifier circuitry and operable to apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes and obtain the secured set of bandpass components based on the secured set of mapping codes.

The previous detailed description is merely illustrative in nature and is not intended to limit the present disclosure, or the application and uses of the present disclosure. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding field of use, background, summary, or detailed description. The present disclosure provides various examples, embodiments and the like, which may be described herein in terms of functional or logical block elements. The various aspects described herein are presented as methods, devices (or apparatus), systems, or articles of manufacture that may include a number of components, elements, members, modules, nodes, peripherals, or the like. Further, these methods, devices, systems, or articles of manufacture may include or not include additional components, elements, members, modules, nodes, peripherals, or the like.

Furthermore, the various aspects described herein may be implemented using standard programming or engineering techniques to produce software, firmware, hardware (e.g., circuitry), or any combination thereof to control a computing device to implement the disclosed subject matter. It will be appreciated that some embodiments may be comprised of one or more generic or specialized processors such as

microprocessors, digital signal processors, customized processors and FPGAs and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuitry, some, most, or all of the functions of the methods, devices and systems described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more ASICs, in which each function or some combinations of certain of the functions are implemented as custom logic circuitry. Of course, a combination of the two approaches may be used. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

The term "article of manufacture" as used herein is intended to encompass a computer program accessible from any computing device, carrier, or media. For example, a computer-readable medium may include: a magnetic storage device such as a hard disk, a floppy disk or a magnetic strip; an optical disk such as a CD or DVD; a smart card; and a flash memory device such as a card, stick or key drive. Additionally, it should be appreciated that a carrier wave may be employed to carry computer-readable electronic data including those used in transmitting and receiving electronic data such as electronic mail (e-mail) or in accessing a computer network such as the Internet or a LAN. Of course, a person of ordinary skill in the art will recognize many modifications may be made to this configuration without departing from the scope or spirit of the subject matter of this disclosure.

Throughout the specification and the embodiments, the following terms take at least the meanings explicitly associated herein, unless the context clearly dictates otherwise. Relational terms such as "first" and "second," and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The term "or" is intended to mean an inclusive "or" unless specified otherwise or clear from the context to be directed to an exclusive form. Further, the terms "a," "an," and "the" are intended to mean one or more unless specified otherwise or clear from the context to be directed to a singular form. The term "include" and its various forms are intended to mean including but not limited to. References to "one embodiment," "an embodiment," "example embodiment," "various embodiments," and other like terms indicate that the embodiments of the disclosed technology so described may include a particular function, feature, structure, or characteristic, but not every embodiment necessarily includes the particular function, feature, structure, or characteristic. Further, repeated use of the phrase "in one embodiment" does not necessarily refer to the same embodiment, although it may. The terms "substantially," "essentially," "approximately," "about" or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. A device or structure that is "configured" in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

What is claimed is:

1. A method, comprising:

by an electronic device having a circuitry electrically coupled to a set of amplifier circuitry operable to collectively amplify an input signal having information,

outputting, by the circuitry, a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components, with the secured set of bandpass components collectively representing a sample of the input signal and corresponding to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

2. The method of claim 1, wherein each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample.

3. The method of claim 1, wherein at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

4. The method of claim 1, further comprising:

applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and

obtaining the secured set of bandpass components based on the secured set of mapping codes.

5. The method of claim 1, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

6. The method of claim 1, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

7. The method of claim 1, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

8. The method of claim 1, wherein each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

9. An electronic device, comprising:

a set of amplifier circuitry operable to collectively amplify an input signal having information; and

a circuitry electrically coupled to the set of amplifier circuitry and operable to output a set of bandpass components that is secured by a physical layer security scheme to obtain a secured set of bandpass components, with the secured set of bandpass components collectively representing a sample of the input signal and corresponding to both the set of amplifier circuitry and a quantized polar representation of the input signal sample.

10. The device of claim 9, wherein each bandpass component is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

11. The device of claim 9, wherein at least one amplifier circuitry is configured to output power that is linearly proportional to another of the set of amplifier circuitry or at

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least one amplifier circuitry is configured to output power that is non-linearly proportional to another of the set of amplifier circuitry.

12. The device of claim 9, wherein the first circuitry is further operable to:

apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and

obtain the secured set of bandpass components based on the secured set of mapping codes.

13. The device of claim 9, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

14. The device of claim 9, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

15. The device of claim 9, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

16. The device of claim 9, wherein each bandpass component corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

17. A method, comprising:

by an electronic device having a set of amplifier circuitry, amplifying, by the set of amplifier circuitry, a set of output signals that collectively represents an amplified sample of an input signal having information, with the set of output signals corresponding to a set of bandpass components that is secured by a physical layer security scheme and associated with a quantized polar representation of the input signal sample.

18. The method of claim 17, wherein each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

19. The method of claim 17, further comprising:

applying the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and

obtaining the secured set of bandpass components based on the secured set of mapping codes.

20. The method of claim 17, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

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21. The method of claim 17, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

22. The method of claim 17, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

23. The method of claim 17, wherein each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

24. An electronic device, comprising:

a set of amplifier circuitry operable to output a set of output signals that collectively represents an amplified sample of an input signal having information, with the set of output signals corresponding to a set of bandpass components that is secured by a physical layer security scheme and associated with a quantized polar representation of the input signal sample.

25. The device of claim 24, wherein each output signal is associated with one of a set of quantization bits that represents a quantized amplitude of the polar representation of the sample.

26. The device of claim 24, wherein the device further includes a first circuitry electrically coupled to the set of amplifier circuitry and operable to:

apply the physical layer security scheme to a set of mapping codes based on the quantized polar representation of the input signal sample to obtain a secured set of mapping codes; and

obtain the secured set of bandpass components based on the secured set of mapping codes.

27. The device of claim 24, wherein the physical layer security scheme is associated with a permutation of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

28. The device of claim 24, wherein the physical layer security scheme is associated with a puncture of at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

29. The device of claim 24, wherein the physical layer security scheme is associated with a scale factor or a phase rotation applied to at least one of a set of mapping codes that corresponds to the secured set of bandpass components.

30. The device of claim 24, wherein each output signal corresponds to one of the set of amplifier circuitry, one of a set of quantization bits that represents a quantized amplitude of the polar representation of the input signal sample, and a phase of the polar representation of the input signal sample.

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